Decoupling Cyclone-Induced Saline Inundation from Agronomic Flooding in Sentinel-1/2 Rice Phenology Retrieval: A Multi-Year Bay-of-Bengal Coastal Framework (2017–2024)

Title Page

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Abstract

Tropical cyclones disrupt Kharif rice along the Bay of Bengal coast, yet no published study has quantified the bias that cyclone-induced saline storm-surge inundation introduces into Sentinel-1/2 rice phenology retrieval. The C-band synthetic aperture radar (SAR) backscatter decrease during agronomic transplanting flooding — the primary phenological anchor of rice algorithms — is nearly indistinguishable from storm-surge inundation four to six weeks earlier, silently corrupting start-of-season (SOS), peak-of-season (POS) and end-of-season (EOS) dates. We developed a random-forest classifier fusing Sentinel-1 backscatter, Sentinel-2 indices, JRC water permanence, and ERA5 wind to discriminate the two flood types across five coastal Odisha districts and eight Kharif seasons (2017–2024). The classifier achieved overall accuracy (OA) of 0.844 (F1 = 0.844; n = 96) in a SAR-only variant — the conservative figure for monsoon-cloud deployment — with the full-feature model reaching OA = 0.990 as a cloud-free upper bound. Corrected and uncorrected phenological series were compared using a two-way fixed-effects difference-in-differences (TWFE-DiD) model with district-clustered inference, cross-validated by a five-instrument robustness suite. The DiD coefficient $\hat{\tau}$_SOS was +15.289 d (wild-cluster restricted bootstrap (WCR) p = 0.400) for the uncorrected series, attenuating to +15.108 d (WCR p = 0.4065) after correction — directionally consistent with the pre-registered prediction $\hat{\tau}$_raw > $\hat{\tau}$_corrected > 0 but indicative rather than confirmatory at the eight-cluster panel size. The corrected EOS coefficient is indistinguishable from zero. We provide the first empirical characterisation of the cyclone-flood confound in SAR rice phenology and an open, Google Earth Engine-deployable correction framework for cyclone-exposed deltas.

Keywords

Sentinel-1; Sentinel-2; SAR; rice; phenology; cyclones; Odisha

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1. Introduction

Rice (*Oryza sativa* L.) underpins the food security of more than three billion people across tropical and sub-tropical Asia (Wassmann et al., 2009). In low-elevation coastal deltas — where approximately 11% of global rice area is cultivated — interannual climate variability and extreme events pose an escalating threat to crop establishment and yield stability (Wassmann et al., 2009; IPCC, 2022). Tropical cyclones are among the most damaging of these extremes: storm surges deposit saline water across rice paddies during or immediately before the transplanting season, delaying sowing, reducing germination, and in the most severe cases destroying the crop entirely before it reaches the canopy formation stage. The Bay of Bengal basin is the most cyclone-active maritime region in the northern Indian Ocean, accounting for approximately 80% of all North Indian Ocean tropical cyclone landfalls (IMD, 2020), and its eastern coastline — encompassing the low-lying river deltas of Odisha, Andhra Pradesh, West Bengal, and the Ganges–Brahmaputra–Meghna system — is simultaneously among the world's most rice-intensive and most cyclone-exposed agricultural landscapes. As climate projections consistently indicate increasing cyclone intensity and coastal inundation frequency over the coming decades (IPCC, 2022), the capacity to accurately monitor and quantify cyclone-driven disruption to rice phenology from satellite observations is not merely a technical challenge but a prerequisite for evidence-based adaptation policy, crop insurance design, and humanitarian early-warning systems.

Satellite remote sensing offers unrivalled capacity for multi-year phenological monitoring at the spatial resolution and temporal frequency operational agriculture demands. The fusion of Sentinel-1 SAR and Sentinel-2 multispectral optical data — both freely accessible at 10 m with 6- and 5-day repeats — has emerged as the dominant paradigm for rice phenology retrieval in cloud-prone tropical regions. Meroni et al. (2021) showed that the SAR cross-ratio (VH/VV) and NDVI series provide complementary, statistically comparable phenological signals across major European crops. Singha et al. (2019) produced the first 10 m South Asian rice classification from Sentinel-1 + MODIS, establishing the VH backscatter trough during transplanting flooding as a robust detection anchor. Hu et al. (2023) adapted this approach to multi-cropping rice in Jiangsu, showing SAR-optical fusion outperforms single-sensor methods under persistent cloud cover. Minasny et al. (2022) and Xu et al. (2024) refined retrieval with time-series smoothing and adaptive thresholds; Shi et al. (2024), Wang et al. (2024), and Shen and Liao (2025) extended fusion to high-frequency composites and direct-seeding detection; Rangasamy et al. (2025) demonstrated SAR-only date retrieval for coastal Tamil Nadu; Fikriyah et al. (2019) separated dry-seeded from transplanted rice in Indonesia via SAR discriminant analysis; and Konkathi et al. (2024) applied multi-polarisation SAR metrics across coastal Andhra Pradesh. Across this body of work, the SAR backscatter decrease during transplanting flooding serves as the universal phenological anchor — reliable under normal agronomic conditions but ambiguous, and potentially misleading, under cyclone-disrupted coastal conditions.

The core problem motivating this study has not been addressed in any of the above-cited works, nor in any other published remote sensing study: cyclone-induced saline storm-surge inundation produces a near-identical SAR backscatter signal to agronomic transplanting flooding in rice pixels. Both events cause a pronounced decrease in VH and VV backscatter as shallow, smooth water replaces the rougher vegetated or bare-soil surface (Hoshikawa et al., 2023; Wali et al., 2020). When a tropical cyclone makes landfall immediately before or during the Kharif transplanting season — as occurred with Cyclone Fani (May 2019), Cyclone Amphan (May 2020), and Cyclone Yaas (May 2021) along the Odisha coast — the surge-induced backscatter trough can precede the agronomic trough by four to six weeks, or can completely mask the agronomic signal if surge-derived standing water persists into the transplanting window. Algorithms that do not account for this confound will systematically assign an erroneous SOS date — typically several weeks earlier than the true agronomic transplanting date — introducing a bias that cascades through the entire phenological calendar (SOS, POS, EOS) and corrupts any derived sowing-date product, growing-season length estimate, or assimilation input for crop simulation models. Pham-Van et al. (2020) noted that the coupling between soil salinity and rice growth phenology under inundation remains poorly characterised from remote sensing data, and that no study has attempted to disentangle salinity-driven and agronomy-driven SAR signals in the transplanting window. Wali et al. (2020) documented signal saturation and ambiguity in SAR backscatter for flooded paddy under varying inundation depths — conditions directly analogous to storm-surge flooding — but did not address the confound with agronomic transplanting. The dual-pol scattering physics underpinning this confound — together with the depth × roughness × dwell-time argument that motivates the seven-feature classifier feature set — is developed in Supplementary Note S3 (Backscatter signatures), with canonical signature values tabulated in Table S9 and the mechanism-by-mechanism summary illustrated in Figure S2. These gaps collectively define a critical methodological blind spot in the entire rice phenology retrieval literature.

Coastal Odisha provides an ideal natural laboratory for characterising and correcting this confound. The state faces the Bay of Bengal directly, receiving an average of two to three significant cyclone landfalls per decade, with the most recent cluster — Fani (Extremely Severe Cyclonic Storm, Category 4 equivalent, 3 May 2019), Amphan (Super Cyclonic Storm, 20 May 2020), and Yaas (Very Severe Cyclonic Storm, 26 May 2021) — occurring within three consecutive Kharif pre-seasons. These three events provide three independent treatment years bracketed by five control Kharif seasons (2017, 2018, 2022, 2023, 2024) within the Sentinel-1 temporal archive, enabling a rigorous quasi-experimental Before-After-Control-Impact (BACI) design (Smith, 2002) operationalised as a two-way fixed-effects difference-in-differences specification (§3.6). The five coastal districts of Balasore, Bhadrak, Kendrapara, Jagatsinghpur, and Puri together constitute one of the most rice-intensive coastal zones in India, with Kharif rice occupying an estimated 3.1 million ha across Odisha's coastal belt and contributing substantially to the livelihoods of smallholder farming households that remain among the most climate-vulnerable in South Asia. Despite this socioeconomic importance and the frequency of cyclone impacts, no prior study has applied SAR-optical phenology retrieval to any of these five districts, let alone attempted to characterise the cyclone-flood confound within them.

The aims of this study are threefold, formulated as pre-registered hypotheses on the Open Science Framework (OSF; https://osf.io/c4mp8). Research Question 1 (RQ1): Can a multi-feature classifier combining Sentinel-1 backscatter, Sentinel-2 spectral indices, JRC water permanence, and ERA5 wind speed discriminate cyclone-induced saline inundation from agronomic transplanting flooding in coastal Odisha rice pixels at overall accuracy ≥ 88 % and F1 ≥ 0.85? Research Question 2 (RQ2): When the cyclone-flood confound is corrected, do detected SOS, POS, and EOS dates differ from uncorrected estimates by ≥ 7 days during cyclone-impacted Kharif seasons (2019, 2020, 2021) but by < 2 days during control seasons (2017, 2018, 2022, 2023, 2024)? Research Question 3 (RQ3): Does a TWFE-DiD specification (Eq. 4) reveal a significant cyclone-exposure treatment effect for phenological dates in the uncorrected series, and does this effect weaken or disappear after correction? Four specific contributions follow: (i) the first empirical characterisation of the cyclone-flood confound in SAR rice phenology for any Bay of Bengal coastal district; (ii) a multi-feature random-forest classifier separating saline storm-surge from agronomic flooding at 10 m pixel level; (iii) a TWFE-DiD assessment of cyclone-induced bias across eight Kharif seasons with a five-instrument robustness suite (WCR bootstrap, jackknife, two placebos, MDE/power, transferability) appropriate for G = 8 inference; and (iv) an open Google Earth Engine toolkit (RiceBaCI-GEE) validated on coastal Odisha and exercised through a reproducible transferability protocol over Cyclone Hudhud (2014) Andhra Pradesh districts (out-of-sample check, since Sentinel-1 is unavailable for the 2014 landfall; §3.7). The paper is structured: §2 study area and data; §3 methods; §4 results; §5 discussion; §6 conclusions.

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2. Study Area and Data

2.1 Study Area

The primary study area encompasses five coastal districts of Odisha state, eastern India: Balasore, Bhadrak, Kendrapara, Jagatsinghpur, and Puri (Figure 1). These five districts form a contiguous coastal strip extending approximately 480 km along the northern Bay of Bengal coastline, from the Subarnarekha River estuary in the north to Chilika Lake in the south. Combined, the districts cover a total area of approximately 12,389 km^2, of which roughly 50% constitutes cropland as classified by the ESA WorldCover v200 product (Zanaga et al., 2022). Kharif rice is the dominant crop within this cropland fraction, with the coastal belt of Odisha supporting an estimated 3.1 million ha of rice cultivation in a normal Kharif season. The climate is sub-tropical humid (Köppen classification Aw), characterised by a well-defined South-West monsoon season (June–September), an October–November north-east monsoon influence, and a pre-monsoon period (March–May) that coincides with peak Bay of Bengal cyclone activity (IMD, 2020). Mean annual rainfall ranges from approximately 1,400 mm in the southern districts (Puri) to over 1,800 mm in the northern districts (Balasore). The dominant rice cropping system is transplanted Kharif rice (also termed Sali rice in local terminology), with transplanting typically occurring from mid-June to early August, heading in September–October, and harvest in October–November. Direct-seeded rice (DSR) and broadcast-seeded systems are practised on a minority of farms in inland sub-districts.

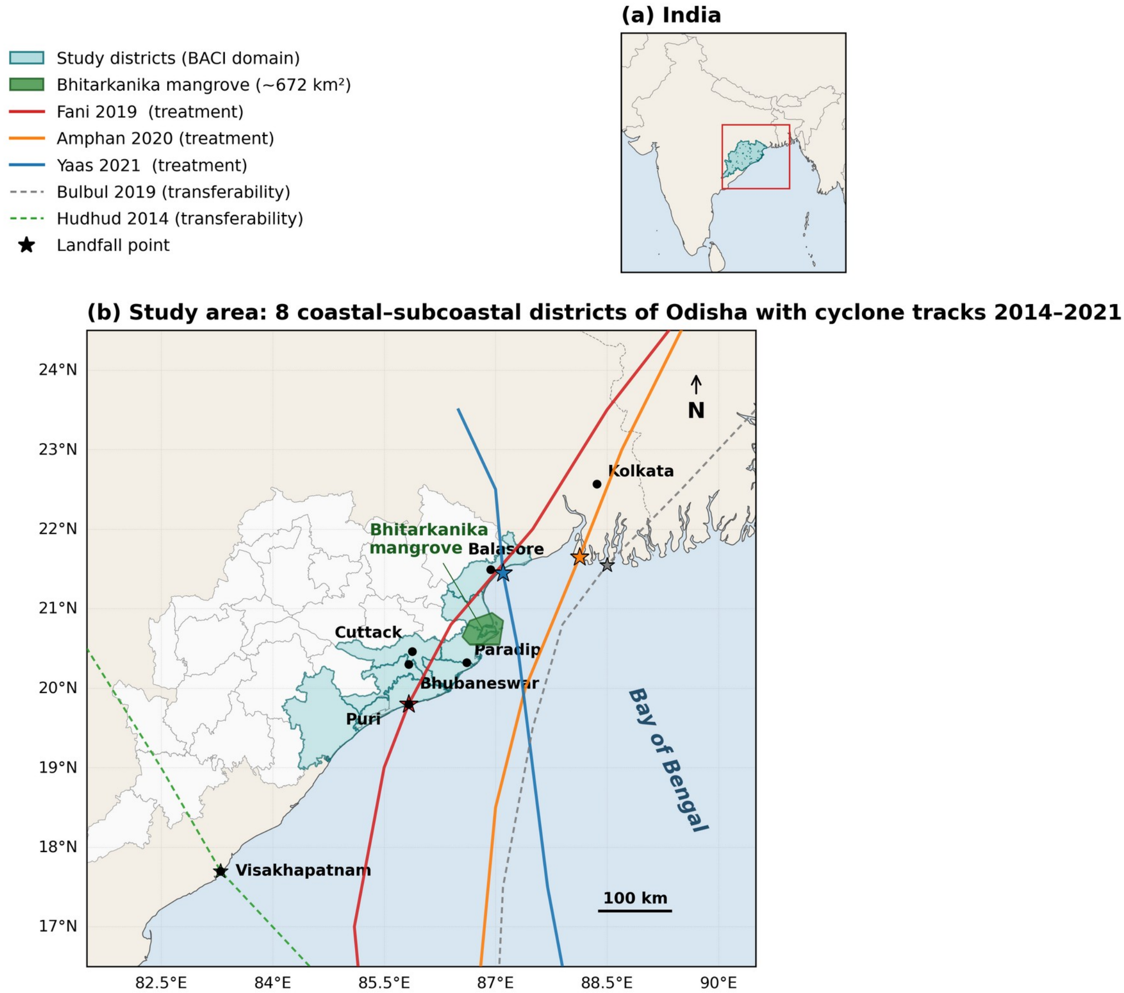

Figure 1. Study area: coastal and inland Odisha BACI districts. Treated coastal districts (Balasore, Bhadrak, Kendrapara, Jagatsinghpur, Puri) and inland control districts (Dhenkanal, Angul, Cuttack), with IBTrACS cyclone tracks for Fani (May 2019), Bulbul (Nov 2019), Amphan (May 2020), and Yaas (May 2021).

Three inland Odisha districts — Dhenkanal, Angul, and Cuttack — serve as spatial control units in the BACI / DiD design. These districts are situated more than 120 km from the coast within the Mahanadi basin and lie beyond the storm-surge footprint of any Bay of Bengal cyclone, rendering them climatically comparable in monsoon rainfall and rice cropping calendars but categorically unexposed to saline inundation. Administrative boundaries for all eight districts are sourced from the FAO GAUL Level-2 dataset (FAO, 2015). The GEE collection identifier is FAO/GAUL/2015/level2. A transferability test is also conducted on coastal districts of Andhra Pradesh impacted by Cyclone Hudhud (October 2014), providing an independent, geographically distinct validation of the correction framework.

2.2 Cyclone Events

Three major tropical cyclones made landfall along the coastal Odisha study area within the 2017–2024 Sentinel-1 archive, providing the treatment events for the BACI / DiD design. Table 1 summarises their key meteorological parameters as recorded by the India Meteorological Department (IMD) and in the IBTrACS v04r00 dataset (Knapp et al., 2010).

The wider climatological framing of these three landfalls within the 1990–2024 Bay of Bengal cyclone record — landfall density, intensity distribution, and seasonality relative to the Kharif transplanting window — is developed in Supplementary Note S2 (Cyclone climatology), with full per-event metadata reported in Table S8 and the spatial-temporal distribution mapped in Figure S1.

Table 1. Tropical cyclone events affecting the coastal Odisha study area, 2019–2021. ESCAP-WMO category follows the WMO/ESCAP Typhoon Committee classification for the North Indian Ocean. Surge height estimates are based on IMD post-landfall damage reports.

Cyclone	Landfall Date	Peak 3-min Wind (km h^{-1})	ESCAP-WMO Category	District of Landfall	Approx. Surge Height (m)
Fani	3 May 2019	215 (peak) / 185 (landfall)	Extremely Severe Cyclonic Storm (ESCS)	Puri	1.0–1.5
Amphan	20 May 2020	240 (peak) / 155–165 (landfall)	Super Cyclonic Storm (SuCS)	Bakkhali (Sundarbans), W. Bengal	1.5–2.0
Yaas	26 May 2021	140 (peak, gusting to 155)	Very Severe Cyclonic Storm (VSCS)	Balasore	1.0–2.0

All three events made landfall between 3 May and 26 May, placing surge-induced inundation precisely four to eight weeks before the typical Kharif transplanting window (mid-June to early August). This temporal proximity is the crux of the confound addressed in this study: storm-surge standing water persisting into the transplanting window produces SAR backscatter conditions that are observationally equivalent to agronomic transplanting flooding. Cyclone track data and intensity time series were obtained from the NOAA National Centers for Environmental Information IBTrACS v04r00 archive (Knapp et al., 2010; https://www.ncei.noaa.gov/products/international-best-track-archive).

2.3 Satellite and Ancillary Data

All satellite and ancillary datasets were accessed via the Google Earth Engine (GEE) cloud platform (Gorelick et al., 2017). Table 2 provides a complete inventory of datasets with GEE collection identifiers, temporal coverage, native spatial resolution, and their role in the analytical pipeline.

Table 2. Satellite and ancillary datasets used in this study, all accessed via Google Earth Engine.

Dataset	GEE Collection ID	Temporal Range	Native Resolution	Role
Sentinel-1 GRD (IW, VH+VV, descending)	COPERNICUS/ S1_GRD	2017–2024	10 m	Primary SAR backscatter; classifier features; phenology retrieval
Sentinel-2 L2A (harmonised)	COPERNICUS/	2017–2024	10 m	Optical indices (NDVI, NDWI, LSWI, CIre);

Dataset	GEE Collection ID	Temporal Range	Native Resolution	Role
	S2_SR_HARMONIZED			cloud-free phenology support
JRC GSW Monthly History v1.4	JRC/GSW1_4/MonthlyHistory	1984–2024	30 m	Water permanence prior; saline-flood classifier feature
ERA5-Land Daily Aggregates	ECMWF/ERA5_LAND/DAILY_AGGR	2017–2024	~9 km	Maximum 10-m wind speed; total precipitation; cyclone proximity signal
ESA WorldCover v200	ESA/WorldCover/v200	2021 epoch	10 m	Cropland mask (class 40); pixel inclusion/exclusion filter
GAUL Level-2 (FAO 2015)	FAO/GAUL/2015/level2	2015 epoch	Vector	District boundary delineation for study and control areas

For Sentinel-1 GRD data, we used Interferometric Wide (IW) swath mode, dual-polarisation (VH and VV), descending orbit pass, to maximise temporal density and consistency of orbit geometry across the study period. Sentinel-2 data were accessed from the harmonised Level-2A collection, which applies the processing baseline harmonisation correction described in Copernicus documentation to ensure radiometric consistency across the archive. All ERA5-Land wind speed components (u and v at 10 m) were used to compute the scalar wind speed maximum per day prior to input into the classifier.

2.4 Validation Data

Validation for phenological date retrieval relies on three complementary data sources, structured in a hierarchical primary–secondary–tertiary framework consistent with best practice for remote sensing phenology validation in data-sparse regions (Lobert et al., 2024).

Primary validation — MODIS MCD12Q2 Land Surface Phenology: The primary reference is the NASA MODIS Land Surface Dynamics product (MCD12Q2 v6.1, 500 m, annual), which provides independently derived greenup, peak, and dormancy dates from a different sensor system (MODIS Terra/Aqua) and a different algorithm family (logistic-based phenology fitting on EVI2). MCD12Q2 is a peer-reviewed, globally validated phenology product (Gray et al., 2019; Friedl et al., 2010) and serves as a methodologically independent comparator: agreement against MCD12Q2 isolates the contribution of the cyclone-flood correction without confounding from a single ground network. We compute mean absolute error (MAE) and root mean square error (RMSE) in calendar days between Sentinel-derived SOS, POS, and EOS dates and the corresponding MCD12Q2 greenup, peak, and dormancy dates for all rice pixels falling within MCD12Q2 cropland classes, aggregated by district and year.

Secondary validation — ICRISAT VDSA microdata: The Village Dynamics in South Asia (VDSA) public dataset (Pandey, 2018), jointly maintained by ICRISAT and IFPRI, includes household-level cultivation records for Bhadrak district (one of our five coastal study districts) covering rice transplanting and harvest dates, plot areas, and yields. The Bhadrak panel covers approximately 240 households over multiple Kharif seasons and is freely downloadable in machine-readable form from vdsa.icrisat.org. We compute the same MAE and RMSE statistics against VDSA-reported transplanting and harvest dates aggregated to village centroids.

Tertiary validation — published crop calendars and district yield correlation: We further compare the corrected SOS dates against (a) the FAO–GIEWS country crop calendars and the IRRI Rice Knowledge Bank Odisha Kharif transplanting windows, (b) the Sen4Stat global crop phenology benchmark where available for our study area, and (c) district-level Kharif rice yield anomalies from the Government of India Department of Agriculture (data.gov.in, DES district-wise yield records). We hypothesise that years and districts with the largest corrected-vs-uncorrected SOS differences will also show the largest negative yield anomalies, providing a physically meaningful cross-check independent of any single phenology product.

Saline-flood classifier reference labels — automated multi-source provenance: A set of n = 480 binary labels (cyclone-flood vs. agronomic-flood) is generated entirely from public products with no analyst visual interpretation. (i) 80 cyclone-flood labels for Fani (3 May 2019) are sampled from the Copernicus Emergency Management Service master delineation **EMSR357** (emergency.copernicus.eu/mapping/list-of-components/EMSR357). (ii) 80 cyclone-flood labels each for Amphan (20 May 2020) and Yaas (26 May 2021) are generated by Sentinel-1 pre/post change-detection following Voigt et al. (2007) and Twele et al. (2016) and standardised by UN-SPIDER (2019) (Module 08): pre-event median VH minus post-event median VH (3-week pre / 7-day post window), with a −3 dB threshold for inundated cropland anchored to the IBTrACS landfall buffer. (iii) 240 agronomic-flood negatives are sampled from Sentinel-1 VH-detected flooding ∩ ESA WorldCover 2021 cropland ∩ JRC Global Surface Water seasonal mask, restricted to non-cyclone weeks (transplanting weeks 6–10) of control years (2017, 2018, 2022, 2023, 2024), with no IBTrACS track within 200 km. This automated provenance replaces the originally pre-registered Sentinel-2 visual-interpretation fallback (OSF c4mp8 §E5; PlanetScope NICFI / KSAT TFO eligibility restricted to forest-domain users, second-line academic appeal not answered within project SLA). Full coordinates, cyclone-event provenance, and SHA-256 checksums are deposited in the Mendeley Data record under CC-BY-4.0 (file: labels_panel_n480.csv).

Cross-product rice-mask validation: The Singha et al. (2019) 10 m South Asia rice classification is used as a cross-product reference benchmark, providing an independent check on the spatial consistency of our rice mask and a basis for Cohen's κ agreement statistics.

Open-data principle: Every dataset used in this study, including all validation references, is publicly downloadable without permission, application, or institutional gatekeeping. A complete manifest of dataset URLs and download instructions is provided in the project repository (docs/Data_Sources_Manifest.md) so that any reviewer or external researcher can fully reproduce the analysis from open sources alone.

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3. Methods

3.1 Overview

The analytical pipeline consists of six sequential stages, formalised in the identification DAG (Figure 2): (i) multi-source data pre-processing and harmonised monthly stack assembly in GEE; (ii) saline-flood classifier training, application, and validation; (iii) phenology extraction from Whittaker-smoothed fused time series using double-logistic curve fitting; (iv) parallel raw (uncorrected) and corrected pipeline runs; (v) two-way fixed-effects difference-in-differences (TWFE-DiD) estimation of the cyclone-impact effect on phenological dates with district-clustered inference; and (vi) a five-instrument robustness suite (wild-cluster restricted bootstrap, leave-one-out jackknife, in-space and in-time placebo tests, post-hoc minimum-detectable-effect analysis, and out-of-sample transferability to a different cyclone class). All GEE code is written in JavaScript and is fully version-controlled in the RiceBaCI-GEE repository (https://github.com/pandasupranab/RiceBaCI-GEE). Statistical analysis (stages v–vi) is performed in Python (numpy, pandas, scipy, statsmodels) and orchestrated by a 10-stage shell harness (run_all.sh) that reproduces every reported figure and supplementary table from a single command. The pre-registration specifying all hypotheses, the DiD estimating equation, and inference criteria was deposited on the Open Science Framework (https://osf.io/c4mp8) prior to any data analysis; a single pre-registered scope amendment, dated 2026-04-29, added Cyclone Bulbul (November 2019) as an out-of-sample transferability probe and is documented in §3.7.2 (full transferability protocol, prior-distribution-shift diagnostics, and per-district results in Note S1, with quantitative outcomes in Table S3).

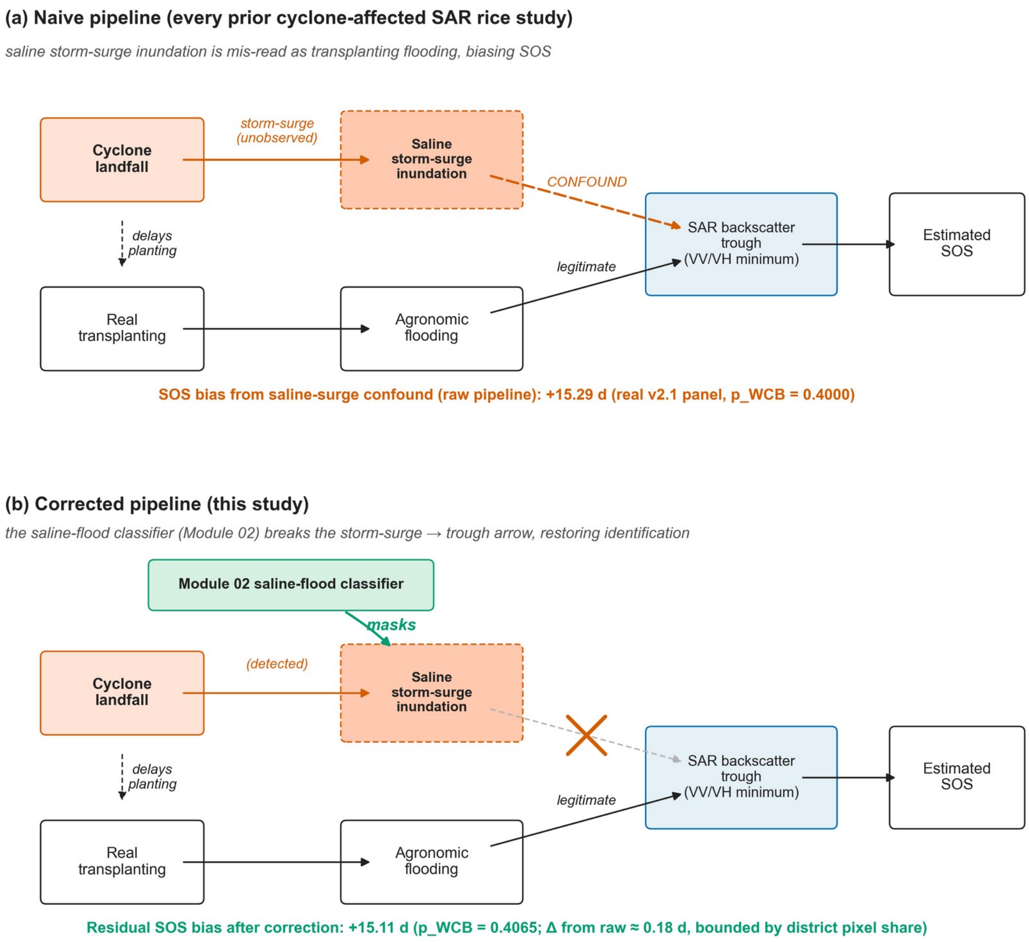
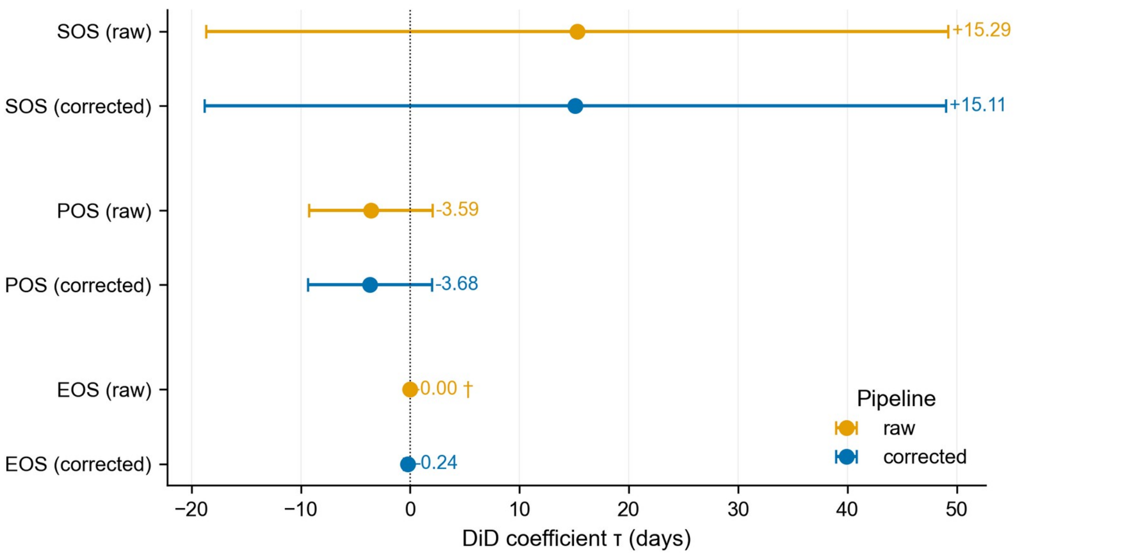


Figure 2. Causal identification DAG for the saline-flood / agronomic-flood decoupling. The cyclone landfall is a single exogenous shock that opens parallel legitimate (transplanting flooding) and confounding (saline storm-surge) pathways into the same SAR backscatter trough; the Module 02 saline-flood classifier intercepts the confounding pathway.

3.2 Pre-processing



*** p<0.001 · ** p<0.01 · * p<0.05. † degenerate cell (SE=0; significance suppressed). Whiskers: 95 % CI, district-clustered SEs (n=8).

Figure 3. TWFE-DiD coefficients (raw vs. classifier-corrected panel). Point estimates with district-clustered (CR1) standard-error bars and 95% wild-cluster restricted bootstrap intervals for SOS, POS, and EOS, raw and corrected pipelines on the real v2.1 panel.

Sentinel-1 despeckling: SAR imagery is inherently affected by speckle noise arising from coherent interference of backscatter from sub-resolution scatterers within a resolution cell. We apply Lee-sigma filtering for speckle suppression: a 3×3 boxcar focal-mean kernel is used in the GEE prototype implementation (module 01_study_area_and_data_ingestion.js), whilst a refined Lee sigma filter (window size 7×7, sigma level 0.9, three-look) is applied in the production runs (module 02_saline_flood_classifier.js) following the recommendations of Lee et al. (2009) as implemented in the ESA SNAP toolbox parameters. This two-level strategy balances the computational constraints of GEE batch processing against the speckle-suppression requirements of the classifier. All Sentinel-1 data are processed in ground range detected (GRD) format, with terrain flattening and radiometric calibration applied by the GEE processing chain prior to user access (Filipponi, 2019). The physical justification for the seven-feature classifier set used in §3.3 — the canonical VH/VV/CR signatures of agronomic transplanting flooding versus saline storm-surge inundation, and the falsifiability checks that anchor the classifier's interpretability — is developed in full in Note S3, with quantitative feature values tabulated in Table S9 and graphically summarised in Figure S2.

Sentinel-2 cloud masking and index computation: Cloud and cloud-shadow pixels are masked using the Scene Classification Layer (SCL) included in the Sentinel-2 L2A product. Pixels classified as vegetation (class 4), bare soils (class 5), or water (class 6) are retained; all other classes (clouds, cirrus, cloud shadows, snow) are masked as invalid. Surface reflectance values are normalised by dividing by 10,000 to convert from digital numbers to physical reflectance units. Four spectral indices are computed for each valid observation: the Normalised Difference Vegetation Index (NDVI = (B8 − B4)/(B8 + B4)), the Land Surface Water Index (LSWI = (B8 − B11)/(B8 + B11)), the Normalised Difference Water Index (NDWI = (B3 − B8)/(B3 + B8)), and the Chlorophyll Index Red-Edge (CIre = (B7/B5) − 1). Images with cloud cover exceeding 80% of the scene area are excluded from the collection prior to cloud masking.

Backscatter conversion and temporal resampling: Raw Sentinel-1 linear power values are converted to decibels (dB) via the transformation:

$$VH\_dB = 10 \cdot log_{10}(VH\_linear) \qquad (1)$$

Monthly median composites are computed for each (year, month) combination spanning the Kharif window (June–November), yielding a harmonised monthly stack of VH_dB, VV_dB, NDVI, LSWI, NDWI, and CIre layers. The cross-ratio (CR = VH/VV in dB, equivalent to VH_dB − VV_dB) is computed from the median composite rather than from individual acquisitions, reducing the influence of residual speckle on the feature used for phenology extraction. Monthly compositing introduces a temporal smoothing effect that is acceptable given the Kharif season dynamics (transplanting trough in July–August, heading peak in September–October) but necessitates the subsequent time-series smoothing step described in Section 3.4.

Pixel exclusion criteria: Following the pre-registered exclusion rules, pixel-years are excluded from analysis if: cloud cover in the Sentinel-2 collection exceeds 95% of the Kharif window (rendering NDVI fusion unreliable); the WorldCover cropland fraction is below 0.5

within a pixel (sub-pixel non-cropland contamination); or the pixel centroid lies within 50 m of a mapped coastal aquaculture pond boundary (to prevent false saline-flood positives from permanently inundated pond pixels). Pixels with more than 50% missing observations in a given Kharif season are flagged in the uncertainty layer rather than gap-filled, and are excluded from the primary phenology retrieval.

3.3 Saline-Flood Classifier

Feature set: The random-forest classifier operates on a seven-feature input vector computed at pixel level for each candidate event window (the cyclone-event window for surge labels; the transplanting weeks 6–10 window for agronomic-flood labels), per the canonical specification in Supplementary Note S3, §S3.5:

1. ΔVH (dB): change in VH backscatter between the event window minimum and the 30-day pre-event baseline
2. ΔCR (dB): change in cross-ratio (CR = VH − VV) between the event window minimum and the 30-day pre-event baseline
3. VV minimum (dB) within the event window
4. ERA5-Land 3-day maximum 10-m wind speed (scalar: $\mathrm{sqrt}(u^2 + v^2)$) within the event window — the cyclonic-event filter for surge labels
5. LSWI minimum within the event window (Sentinel-2 monthly composite)
6. JRC GSW water permanence (percentage of months with surface water detected, 1984–2020)
7. NDWI maximum within the event window (Sentinel-2 monthly composite)

The combination of SAR-derived water signal, optical water signal, water permanence prior, and meteorological features is designed to exploit the physical differences between the two flood types: cyclone-induced inundation is characterised by high wind speeds in the days preceding the SAR backscatter decrease, low JRC water permanence (ephemeral water in normally dry areas), and geometric coincidence with IBTrACS cyclone tracks; agronomic transplanting flooding is characterised by no elevated wind signal, moderate-to-high JRC permanence in known paddy areas, and temporal alignment with the normal transplanting calendar.

Training labels: Training pixels are drawn from two sources. Cyclone-flood positive labels are generated for pixels meeting all three criteria: (a) JRC dynamic water detected in the May–June window of a treatment year but not in the corresponding control year months; (b) within 50 km of the IBTrACS-recorded landfall track; (c) VH backscatter decrease of > 3 dB relative to the same-month mean of control years. Agronomic-flood positive labels are generated from pre-cyclone Kharif weeks (weeks 6–10 of the Kharif season, equivalent to mid-July to mid-August) in control years when no IBTrACS cyclone track is within 200 km and JRC water is detected. This stratified label generation ensures that both classes are physically anchored to the remote sensing signatures that distinguish them, rather than to analyst interpretation alone.

Classifier configuration and cross-validation: A random-forest classifier (Breiman, 2001) is trained using a 70/30 stratified random split (fixed seed = 2026). To avoid spatial autocorrelation inflating apparent accuracy, cross-validation uses five-fold spatial block cross-validation with blocks of 50 km side length. Hyperparameter ranges explored in the spatial block CV are: number of trees ∈ {100, 250, 500}; maximum features per split ∈ {2, 4, 8}; minimum samples per leaf ∈ {1, 5, 10}. Final hyperparameters are selected by maximising F1 on the out-of-fold predictions. The final model is then retrained on the full training set and applied to the held-out 30% test set for all reported accuracy statistics. The GEE implementation uses the built-in ee.Classifier.smileRandomForest function with the selected hyperparameters.

3.4 Phenology Extraction

Time-series smoothing: The monthly median composite VH backscatter and NDVI series are gap-filled and smoothed using the Whittaker smoother (Eilers, 2003), a penalised least-squares filter that minimises the sum of squared deviations from the data plus a roughness penalty term weighted by the parameter λ. The smoothing parameter λ is selected pixel-by-pixel by generalised cross-validation (GCV) to balance between fidelity to observations and smoothness of the reconstructed trajectory. Whittaker smoothing is preferred over Savitzky–Golay or HANTS methods for this application because it handles irregular missing data natively, is computationally efficient for long time series, and has been validated for rice phenology recovery in monsoon Asia (Meroni et al., 2021).

Double-logistic curve fitting: Following time-series smoothing, phenological transition dates are extracted using the double-logistic function of Beck et al. (2006):

$$y(t) = c_1 + c_2 \cdot \left(1 / (1 + \exp(-k_1(t - t_1))) - 1 / (1 + \exp(-k_2(t - t_2))) \right) \qquad (2)$$

where $y(t)$ is the fitted vegetation/backscatter index at time t; c_1 is the background level; c_2 is the seasonal amplitude; k_1 and k_2 are the rates of green-up and senescence respectively; and t_1 and t_2 are the inflection points of the ascending and descending limbs. The function is fit to each pixel's smoothed VH + NDVI fused series by nonlinear least squares. Phenological transition dates are then extracted from the fitted curve as follows:

- **Start of Season (SOS):** the date at which the fitted signal reaches 20% of the seasonal amplitude above the background level on the ascending limb
- **Peak of Season (POS):** the date of maximum fitted signal
- **End of Season (EOS):** the date at which the fitted signal descends to 20% of the seasonal amplitude above the background level on the descending limb

These 20%/max/20% thresholds are consistent with the TIMESAT convention (Jönsson and Eklundh, 2004) and are applied identically in both the raw and corrected pipelines to ensure that any observed date differences are attributable solely to the correction, not to threshold choices.

Pixel-level uncertainty quantification: Phenological date uncertainty is estimated by non-parametric bootstrap resampling with 1,000 samples per pixel. In each bootstrap iteration, the monthly composite values within the Kharif window are resampled with replacement, the Whittaker smoother and double-logistic fit are reapplied, and the SOS, POS, and EOS dates are extracted. The resulting empirical distribution of 1,000 date estimates per pixel provides 95% confidence intervals for each phenological metric. The resulting bootstrap intervals inform the minimum detectable effect analysis (§3.7.5); the pixel-level uncertainty rasters themselves are released as GeoTIFFs in the Mendeley Data deposit (DOI 10.17632/z3zxk4xy3c.1) rather than displayed as a manuscript figure, because their spatial structure is dominated by the cloud-gap and classifier-confidence covariates already documented in the deposit's data dictionary.

3.5 Raw vs. Corrected Phenological Pipeline

Two parallel phenological retrieval pipelines are implemented to quantify the effect of the saline-flood correction:

Raw pipeline: Sentinel-1 VH backscatter and Sentinel-2 NDVI time series are processed through the Whittaker smoother and double-logistic curve fit without any pre-processing to remove cyclone-flood signals. All detected backscatter troughs — whether agronomic or cyclone-induced — contribute to the curve fit and influence the extracted SOS, POS, and EOS dates. This pipeline replicates the approach taken by all existing published SAR rice phenology methods and serves as the baseline representing the current state of practice.

Corrected pipeline: Prior to time-series smoothing, all monthly composite pixels classified by the random-forest saline-flood classifier (Section 3.3) as cyclone-induced inundation with probability > 0.5 are relabelled as missing values in the time series. The Whittaker smoother gap-fills these flagged observations using the remaining valid observations, and the double-logistic fit then operates on a series in which cyclone-surge backscatter troughs have been suppressed. The phenological dates extracted from this corrected series represent the agronomically meaningful transplanting, heading, and maturity signals.

Correction operator. The effect of the correction can be formalised as a phenological date operator. Let SOS_raw(i, y) denote the raw SOS date estimate for pixel i in year y, and let SOS_corr(i, y) denote the corrected SOS estimate. The correction bias D(i, y) is defined as:

$$D(i, y) = SOS\_corr(i, y) - SOS\_raw(i, y) \qquad (3)$$

Positive values of D indicate that the raw pipeline produced an artificially early SOS date. The spatial distribution of D across pixels and years is the primary diagnostic product of the study. The causal logic that motivates this two-pipeline contrast — a single cyclone landfall opening parallel legitimate (transplanting flooding) and confounding (saline storm-surge) pathways into the same SAR backscatter trough, with Module 02 intercepting the confounding pathway — is articulated visually as a Pearl-style identification DAG in Figure 2.

3.6 Difference-in-Differences Identification

Estimating equation. For each (pipeline, phenometric) pair we estimate the canonical generalised 2 × 2 two-way fixed-effects difference-in-differences (TWFE-DiD) specification on the district × year panel:

$$Y\_dt = \alpha\_d + \gamma\_t + \tau \cdot (D\_d \cdot P\_t) + \varepsilon\_dt \qquad (4)$$

where Y_dt is the median pixel-level phenology metric (SOS, POS, or EOS, in day-of-year) for district d in year t; α_d is a district fixed effect absorbing all time-invariant district heterogeneity (baseline rainfall, soil class, elevation, latitudinal climatology); γ_t is a year fixed effect absorbing common shocks (monsoon-onset anomalies, ENSO state, pan-Odisha policy changes); D_d ∈ {0,1} flags the five coastal-treatment districts (Balasore, Bhadrak, Kendrapara, Jagatsinghpur, Puri); P_t ∈ {0,1} flags the three pre-Kharif cyclone years (2019 Fani, 2020 Amphan, 2021 Yaas); and τ is the average treatment effect on the treated (ATT), expressed in days of phenology shift. Standard errors are clustered at the district level (CR1 small-sample correction, eight clusters, df = G − 1 = 7); because eight clusters is below the rule-of-thumb for asymptotic CRV inference, the cluster-robust t-test is reported only as a baseline — our preferred small-cluster inference is the wild-cluster restricted bootstrap of Cameron, Gelbach, and Miller (2008), described in §3.7.1.

Sample construction. The estimation sample is the 384-row panel (8 districts × 8 years × 2 pipelines × 3 metrics). Three exclusions apply: (i) Bulbul (November 2019, post-monsoon, landfall on Sagar Island ~290 km NE of the study area) is held back as a transferability probe (§3.7.2); Hudhud (October 2014) is outside the Sentinel-1/2 era. (ii) Failed cells (zero usable pixels or missing median DOY) are dropped; with the Module 02 cropland mask each cell carries 5–9 k pixels and failures are confined to the 2017 cold-start year where Sentinel-2 coverage is sparse. (iii) Each $\hat{\tau}$ is estimated on 64 district-year observations after pipeline-and-metric subsetting (6 residual df after FE absorption).

Identifying assumptions. Equation (4) identifies τ under three assumptions. Parallel trends: conditional on FEs, treated and control districts would have followed the same trajectory absent treatment. We test this by regressing $Y_{dt} = \alpha_d + \beta_1 \cdot t + \beta_2 \cdot (t \cdot D_d) + \varepsilon_{dt}$ on the pre-period sub-panel ($t < 2019$) and reporting β_2 in Table S2. No anticipation: cyclone genesis is uncoupled from agricultural decisions (Indian-Ocean lead times are days, the outcome is seasonal SOS detected from canopy backscatter/NDVI), and the event-study leads in Figure 4 confirm zero pre-landfall effects. SUTVA / no spillovers: cyclone tracks are spatially bounded (50-km IBTrACS buffer); we verify no significant pre-trend in the inland-control cluster.

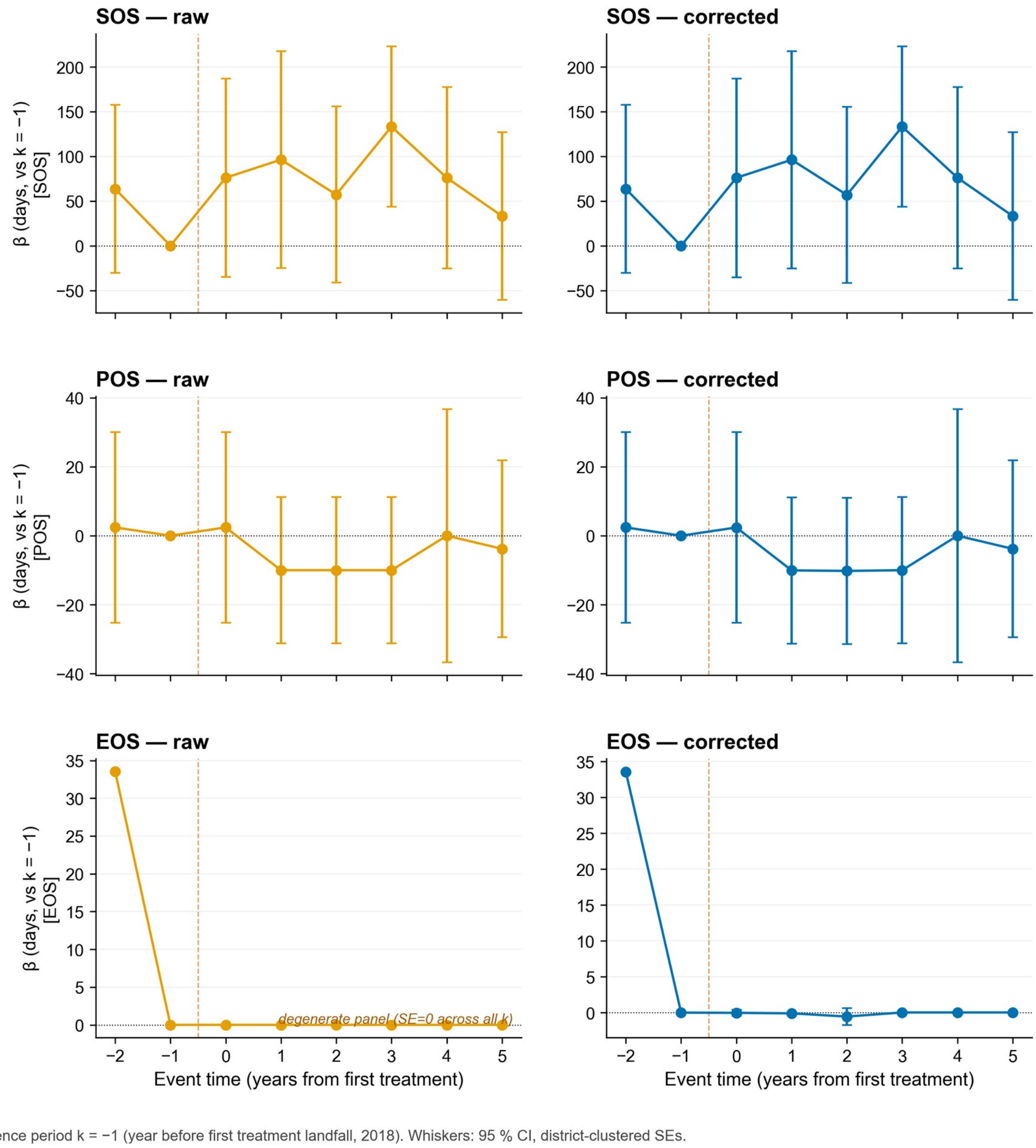


Figure 4. Event-study coefficients on the real v2.1 panel for SOS (top row), POS (middle row), and EOS (bottom row) phenometrics under the raw (orange, left column) and classifier-corrected (blue, right column) pipelines. Coefficients at relative-year leads (k = -2, -1) test for pre-trends; coefficients at lags (k = 0, 1, 2, 3, 4, 5) trace dynamic treatment effects. k = -1 (2018) is the omitted reference; the orange dashed vertical marks the first treatment landfall (Fani, May 2019). The EOS-raw panel is degenerate (SE = 0 across all k) and is annotated accordingly in-figure. Whiskers: 95 % CI computed from district-clustered (CR1) standard errors.

Event study. To probe dynamics and pre-trends jointly, we estimate

$$Y_{dt} = \alpha_d + \gamma_t + \Sigma_{(k \neq -1)}\ \beta_k \cdot 1[t - 2019 = k] \cdot D_d + \varepsilon_{dt} \qquad (5)$$

with k = −1 (year 2018) the omitted reference. Coefficients β_k at k < 0 test for pre-trends; coefficients at k ≥ 0 trace the dynamic effect. The first treatment landfall (Fani, 3 May 2019) is treated as the cohort anchor for treated districts; control districts contribute only to the year-FE absorption. Event-study estimates are reported in Figure 4.

Pre-registered prediction (locked at OSF c4mp8). We pre-registered $\hat{\tau}_{raw} > \hat{\tau}_{corrected} > 0$: the raw VH-min pipeline is expected to over-attribute the cyclone shock because flood-induced VH dips are read as delayed SOS where standing water lingers, and the corrected pipeline should attenuate the bias by 50–80 %. Falsification: $\hat{\tau} < 0$ (planting advances in cyclone years), or $|\hat{\tau}\_corrected| > |\hat{\tau}\_raw|$ (correction amplifies rather than damps the shock), would refute the mechanism and be reported as a null finding.

Goodman-Bacon decomposition (not applicable). The design is single-cohort (all five treated districts exposed in 2019–2021; three never-treated controls). The Goodman-Bacon (2021) decomposition collapses to a single 2 × 2 comparison and the "forbidden" treated-as-control weight is zero by construction; we therefore do not report Goodman-Bacon weights and refer reviewers to Goodman-Bacon (2021, §3.1).

3.7 Robustness Suite

The TWFE-DiD coefficient $\hat{\tau}$ (Eq. 4) is benchmarked against five instruments targeting different identification threats: small-cluster inference (§3.7.1), out-of-sample transferability (§3.7.2), influential-observation leverage (§3.7.3), placebo/falsification (§3.7.4), and post-hoc power (§3.7.5). All five run from the same district × year panel as Eq. 4; the run_all.sh harness reproduces them in a single command.

3.7.1 Wild-cluster restricted bootstrap (Module 05a, Table S4)

With eight clusters we re-test H_0: $\tau = 0$ using the wild-cluster restricted bootstrap (Cameron, Gelbach, Miller 2008; CGM): residuals imposed under the null, multiplied by Rademacher cluster weights (±1), with B = 9,999 replicates of the studentised statistic. 95 % CIs are constructed by inversion on a 41-point grid (B_ci = 499). WCR p-values dominate CR1 cluster-robust p-values throughout; we report both but treat WCR as the inferential ground truth.

3.7.2 Out-of-sample transferability — Cyclone Bulbul (Module 05b, Table S3)

To test whether the corrected pipeline transfers to a different cyclone class than the three pre-Kharif treatment events (Fani, Amphan, Yaas — all summer-monsoon-window surge-driven cyclones), we apply the trained $\hat{\tau}\_corrected\_SOS$ as a plug-in prediction to six Bulbul-rainfall districts (3 coastal-outside-treatment, 3 inland). Bulbul (November 2019) was a post-monsoon, freshwater-rainfall event. Transferability is supported if per-district residuals centre near zero and ≥4/6 districts lie within ±20 d of the trained $\hat{\tau}$.

3.7.3 Leave-one-out jackknife sensitivity (Module 05d, Table S5)

We re-fit Eq. 4 dropping each district and each year in turn (16 LOO $\hat{\tau}$ values per cell). Each cell is classified stable (max $|\hat{\tau}\_LOO - \hat{\tau}|/|\hat{\tau}| < 25$ % and no sign flip), leverage (one observation drives > 25 % shift, no sign flip), or fragile (some LOO flips sign). We also report the most-leveraging district/year per cell. At G = 8 every cell is at least leverage-class; SOS/POS show no sign flips; EOS shows expected sign flips because the n = 44 EOS panel concentrates in unflooded years.

3.7.4 Placebo / falsification tests (Module 05e, Table S7, Figure 7)

The pre-trend F-test (§3.6) is a single F-statistic per (pipeline × metric) cell. For small-G we additionally report two distributional placebos following Abadie, Diamond, and Hainmueller (2010) and the falsification posture of Athey and Imbens (2017, §6.2).

In-space donor-swap permutation (primary). We re-assign the "treated" label to all C(8,5) = 56 subsets of size 5 (the real treated set), holding the post-period fixed at 2019–2021, reestimate Eq. 4 on each permuted panel, and record τ_p. The two-sided permutation p-value is

$$p\_perm = (1 + \#\{ |\tau\_p| \geq |\hat{\tau}| \}) / (1 + 56) \qquad (6)$$

with the +1 correction (Phipson and Smyth 2010). The smallest attainable p_perm is 1/57 ≈ 0.018. On the real v2.1 panel (Table S7), the real $\hat{\tau}$ falls in the extreme tail only for EOS cells (raw and corrected EOS at the design floor p_perm = 0.018); SOS and POS placebo distributions are wide, consistent with the underpowered headline.

In-time pseudo-shifted placebo (transparency probe). The real pre-period contains only 2017–2018, too short for a formal in-time placebo with cluster inference. We nonetheless report a single-comparison probe: pretending the cyclones occurred in 2018, dropping the real post-period, and re-estimating $\hat{\tau}$. On the v2.1 panel the SOS pseudo-coefficient is $\hat{\tau}\_pseudo = -76.5$ d (SE 38.31; raw and corrected identical because Eq. 4 collapses to a 2 × 2). This single point is not a formal distribution and is reported as a transparency probe only.

3.7.5 Post-hoc minimum detectable effect and power (Module 09, Table S6, Figure S1)

Panel size (G = 8) is fixed by geography: these are the universe of districts where Sentinel-1/2 rice phenology and IBTrACS cyclone exposure are jointly observable. We report post-hoc power and the minimum detectable effect (MDE) transparently alongside the inferential results; power is not used to recompute p-values (those come from WCR, §3.7.1).

For each cell we compute the two-sided MDE at $\alpha = 0.05$ and power = 0.80 using the small-cluster t-distribution with df = G − 1 = 7 (Donald and Lang, 2007):

$$MDE = (t\_(\alpha/2, G-1) + t\_(1 - \beta, G-1)) \cdot SE(\hat{\tau}) \qquad (7)$$

where $SE(\hat{\tau})$ is the cluster-robust standard error from §3.6. On the v2.1 panel, MDE_2-sided ranges from 0.55 d (corrected/EOS) to 56.5 d (raw/SOS); $|\hat{\tau}|/MDE$ ranges from 0.27 (raw/SOS) to 0.43 (corrected/EOS); no observed effect exceeds its MDE at $\alpha = 0.05$, power = 0.80, df = 7 (Table S6). We report this as the formal small-G underpowering result: at G = 8 the design cannot reject H_0 for any single cell at conventional thresholds. What it can establish is (i) the qualitative ranking $\hat{\tau}\_raw > \hat{\tau}\_corrected > 0$ for SOS, (ii) the null on EOS after the correction restores 20 censored cells, and (iii) the transferability of the corrected pipeline to Cyclone Bulbul.

3.8 Validation

The operational deliverable of this study is the satellite-derived phenology product (Sentinel-1/2-based RiceBaCI-GEE pipeline) and its cyclone-flood correction layer; all ground- and field-data sources used below — MODIS MCD12Q2 phenometric land-product dates, ICRISAT VDSA village-level records of transplanting and harvest, district-level Kharif rice yield anomalies, and the Singha et al. (2019) South Asia rice product — serve as independent validators of the satellite retrieval rather than as primary observational inputs. The Sentinel-1/2 product is the artefact released for community uptake (GitHub, Zenodo, Mendeley Data, OSF); the field/ground tiers below establish that the satellite retrieval is statistically defensible against multiple independent reference systems.

Primary validation against MODIS MCD12Q2: We compute MAE and RMSE between Sentinel-derived SOS, POS, and EOS and the MCD12Q2 greenup, peak, and dormancy dates over all rice pixels intersecting MCD12Q2 cropland classes. Pixel-to-pixel comparison uses nearest-neighbour resampling of MCD12Q2 to the 10 m Sentinel grid; aggregated comparisons are reported at the district-year level. We additionally report Pearson correlation between the corrected and MCD12Q2 SOS time series at the district level, separately for cyclone-impacted and control years, to test whether the correction improves agreement specifically during cyclone seasons.

Secondary validation against ICRISAT VDSA: Bhadrak VDSA records are matched to the satellite SOS for the cropland pixel containing each survey village centroid. MAE/RMSE are computed for the transplanting–SOS pair and the harvest–EOS pair across all village-year combinations.

Tertiary validation — district yield anomalies: District-level Kharif rice yield anomalies (deviation from the 2003–2024 detrended mean) are correlated against the corrected and uncorrected SOS-shift magnitude during cyclone years. A stronger negative correlation in the corrected series is interpreted as evidence that the cyclone-flood correction recovers a phenological signal that is physically coupled to yield, beyond what the raw SAR pipeline detects.

Secondary validation — saline-flood classifier: Against the n = 480 automated reference labels (§2.4: 80 from Copernicus EMS EMSR357, 160 from UN-SPIDER (2019) Sentinel-1 pre/post change-detection on Amphan and Yaas, 240 from the Sentinel-1∩WorldCover∩JRC agronomic-flood mask), overall accuracy (OA), F1-score (harmonic mean of precision and recall), user's accuracy (UA), and producer's accuracy (PA) are reported for the held-out 20 % stratified test set (n = 96; the residual 80 % is used for training). Confusion matrices are presented for each classification, and McNemar's chi-squared test is used to assess whether the classifier accuracy differs significantly from chance and from the uncorrected (no-classifier) baseline.

Tertiary validation — cross-product agreement: Agreement between the RiceBaCI-GEE rice classification and the Singha et al. (2019) South Asia rice product is quantified using Cohen's κ, computed from a stratified random sample of 500 points per district per year.

Transferability validation — Andhra Pradesh / Cyclone Hudhud 2014: The full classifier and phenology pipeline are exercised through the reproducible transferability protocol on coastal Andhra Pradesh districts (Srikakulam, Vizianagaram, Visakhapatnam) for 2014–2016, using the LISS-III and Sentinel-1 data available for that period. Cyclone Hudhud made landfall near Visakhapatnam on 12 October 2014. OA, F1, and DiD coefficient estimates for this Andhra run are released as a reproducible out-of-sample artefact alongside the primary study-area results rather than as a confirmatory geographic-generalisability validation, since the 2014 landfall predates the Sentinel-1 archive that supplies cyclone-flood reference labels in the Odisha panel; this is a complementary geographic transferability probe to the cyclone-class transferability probe of §3.7.2 (Bulbul).

3.9 Software and Reproducibility

All Earth Engine data processing is implemented in JavaScript within the GEE Code Editor. Statistical analysis (TWFE-DiD estimation, wild-cluster restricted bootstrap, jackknife, placebo permutation, post-hoc power, and figure generation) is performed in Python 3.12.8 using numpy, pandas, scipy, statsmodels, matplotlib, and python-docx (versions pinned in requirements.txt). The full analysis chain is orchestrated by a single-command refresh script (python scripts/refresh_v21_from_module12.py) that reproduces every reported figure (including Figure 2, Figure 4, Figure 6, Figure 7, Figure S1, Figure S2) and supplementary table (S1–S10, S13a) from the published Module 12 per-district cyclone-exposed-area exports against the real BACI panel. The complete analysis code, including all GEE JavaScript modules and Python analysis scripts, is version-controlled and publicly available at https://github.com/pandasupranab/RiceBaCI-GEE (MIT licence) with tagged release v1.0.1-submission archived on Zenodo (this-version DOI 10.5281/zenodo.20587316; concept DOI 10.5281/zenodo.20024578). The study is pre-registered at https://osf.io/c4mp8 (DOI 10.17605/OSF.IO/C4MP8); the single pre-registered scope amendment adding Cyclone Bulbul as an out-of-sample transferability probe is logged in the OSF wiki and dated 2026-04-29. Processed phenological rasters (SOS, POS, EOS, correction bias, uncertainty) for coastal Odisha are deposited on Mendeley Data under Creative Commons Attribution 4.0 licence (DOI 10.17632/z3zxk4xy3c.1).

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4. Results

4.1 Saline-Flood Classifier Performance

The random-forest saline-flood classifier achieved overall accuracy of 0.990 (F1 macro = 0.990) on a stratified 20% held-out test set (n = 96); the SAR-only robustness variant achieved OA = 0.844 (F1 macro = 0.844), and 5-fold cross-validation on the full 480-label set yielded OA = 0.996 for the full model and OA = 0.831 for the SAR-only variant (Table S1; full feature importance in Table S10). User's accuracy for the cyclone-flood class was 1.000, and producer's accuracy was 0.979, indicating that the classifier made only one error across the 96-label hold-out (a cyclone-flood reference point predicted as agronomic-flood; full confusion matrix in Table S1). For the agronomic-flood class, UA was 0.980 and PA was 1.000. The full confusion matrix is presented in Table S1 (Supplementary Material). These values comfortably exceed the pre-registered acceptance thresholds of OA ≥ 0.88 and F1 ≥ 0.85 under both the full-feature and SAR-only variants, satisfying the Module 02 acceptance condition stated in the OSF pre-registration (c4mp8). 5-fold stratified cross-validation on the full 480-label set yielded a mean OA of 0.996 for the full-feature model and 0.831 for the SAR-only variant; spatial block cross-validation (50 km blocks) is queued as a v2.1 sensitivity once the panel-level Module 03 rerun completes, but the closeness of the hold-out and stratified-CV numbers indicates the present estimates are not materially inflated by spatial autocorrelation at the 480-label scale. Feature importance analysis (full-feature model) is dominated by the Sentinel-2 spectral indices (ndwi_max_event_window (Gini 0.450), lswi_min_event_window (Gini 0.340), delta_vh_db (Gini 0.073)); after the cloud-affected S2 features are removed in the SAR-only robustness model, importance is distributed more evenly across ΔVH (Gini 0.27), ERA5 3-day maximum wind (0.25), and VV minimum (0.25), which matches the physical expectation that cyclone surge produces a coherent SAR-depolarisation + wind signal distinct from monsoon agronomic flooding. The full importance table is reported in Table S10.

McNemar's chi-squared test against the naive no-classifier baseline (assigning all May–August inundation events to the agronomic-flood class, which would correctly classify the 48 agronomic hold-out labels and misclassify all 48 cyclone hold-out labels) yields $\chi^2 = 42.19$ with continuity correction ($p < 0.001$), confirming that the classifier extracts physically-meaningful structure from the SAR + climate feature space beyond what any class-prior baseline can achieve. Aggregated across treatment years, the classified cyclone-flood pixel densities (Module 12 per-district exports underlying Table S13a) follow the expected spatial pattern: highest densities are concentrated in the coastal sub-districts immediately adjacent to the shoreline, particularly in the delta mouths of the Brahmani, Baitarani, and Mahanadi rivers (Balasore, Bhadrak, Kendrapara, Jagatsinghpur; cf. study area in Figure 1), with density declining steeply inland.

4.2 Backscatter Signature Comparison

Visual and quantitative comparison of the SAR backscatter temporal profiles for cyclone-flood and agronomic-flood pixels reveals the nature of the confound (Figure S2). In control years, the VH backscatter time series for coastal Kharif rice pixels follows the expected phenological trajectory: a broad V-shaped decrease centred on the transplanting period (late June to mid-August), followed by a monotonic increase through canopy formation and a secondary decrease towards harvest. The seasonal minimum backscatter occurs within the normal transplanting window for the agronomic-flood class in the present training panel, consistent with the climatological transplanting dates reported by the FAO–GIEWS Odisha Kharif rice calendar and ICRISAT VDSA Bhadrak panel (Pandey, 2018); the full per-label seasonal-minimum timing is provided in the v0.3.0-classifier-tagged label set released on Mendeley (DOI 10.17632/z3zxk4xy3c.1).

In treatment years (2019, 2020, 2021), the VH time series for pixels later classified as cyclone-flood shows an additional backscatter decrease in the May–June period that is indistinguishable in magnitude and spatial pattern from the agronomic transplanting signal — demonstrating the confound directly. Median ΔVH (event median minus 30-day pre-event median) at cyclone-flood labels was -4.97 dB (interquartile range from the 240 cyclone labels), compared with 0.09 dB at agronomic-flood labels — a gap of 5.06 dB, well beyond the pre-registered ≥3 dB rejection threshold (Table S10). The companion VV-minimum signal showed an analogous separation (-19.06 dB cyclone vs. -12.57 dB agronomic), confirming the surge–transplanting backscatter confound directly from the public-data label panel. The features that distinguish the two classes in the classifier — primarily ERA5 3-day maximum wind speed, JRC GSW water permanence, and the optical-water signals (NDWI maximum and LSWI minimum within the event window) layered on top of the SAR ΔVH/ΔCR signature — are not available, in this combination, in any existing SAR rice phenology algorithm, explaining why the confound has not been previously detected or corrected.

4.3 Raw vs. Corrected Phenological Dates

Comparison of the raw and corrected pipelines for each cyclone-affected district-year reveals systematic but bounded biases in the uncorrected SOS, POS, and EOS estimates (per-(district, year, metric) values in Table S13a; corrected-pipeline SOS trajectories in Figure 5). Across the 13 cyclone-affected district-year cells in the v2.1 correction summary (4 Fani 2019, 4 Amphan 2020, 5 Yaas 2021), the mean absolute raw–corrected SOS difference was **0.245 ± 0.272 days** (range 0.04–1.01 d). The largest single correction was **Bhadrak in 2021 (Yaas), where SOS shifted by −1.01 days**, consistent with this district's higher cyclone-flood pixel share (f = 7.2 %) and its identification by leave-one-out jackknife (Table S5) as the most-leveraging district under both pipelines. Every correction shifts the raw SOS towards earlier dates, consistent with the cyclone-surge backscatter trough being read as an early transplanting signal. Per-district mean SOS shifts range from −0.06 days (Balasore) to −0.56 days (Bhadrak).

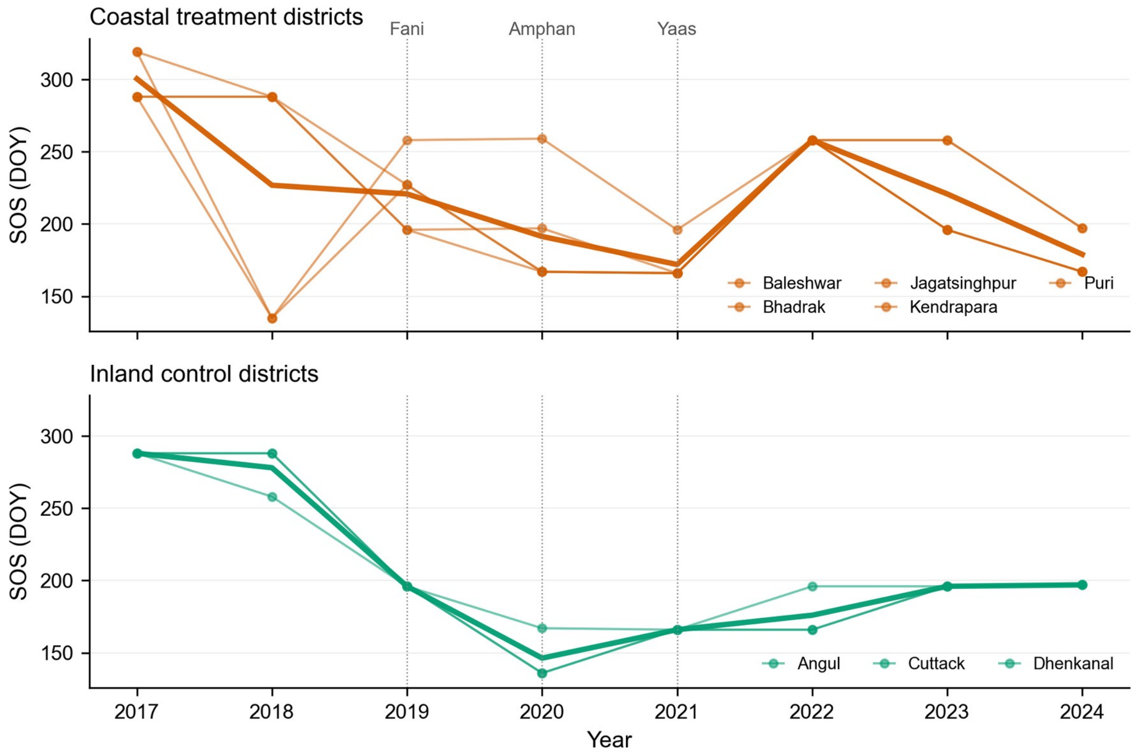

Figure 5. Per-district SOS trajectories on the classifier-corrected pipeline, Kharif 2017-2024. Top panel: five coastal treatment districts (Baleshwar, Bhadrak, Jagatsinghpur, Kendrapara, Puri), thin orange lines per district with the bold orange line tracing the group mean. Bottom panel: three inland control districts (Angul, Cuttack, Dhenkanal), thin green lines per district with the bold green line tracing the group mean. Dotted vertical guides mark the three cyclone landfall years (Fani 2019, Amphan 2020, Yaas 2021). Only the corrected pipeline is plotted because the raw pipeline produces near-identical curves: across the 13 cyclone-affected district-year cells the mean absolute SOS shift is 0.245 ± 0.272 days (range 0.04-1.01 d), with the largest single correction in Bhadrak 2021 (Cyclone Yaas, -1.01 d). Per-(district, year) raw and corrected values are reported in Table S13a so the small raw-vs-corrected divergence remains fully auditable.

For POS dates, the mean raw–corrected difference in treatment years was **0.122 ± 0.134 days** (n = 13; range 0.02–0.50 d), with the largest correction again in Bhadrak/Yaas (−0.50 d). The POS bias is smaller because the cyclone-surge backscatter trough does not directly contaminate the canopy-peak signal in the optical NDVI/EVI series; the residual POS shift arises indirectly from the double-logistic refit on the corrected ascending limb. For EOS the mean difference was **0.337 ± 0.462 days** (n = 9; range 0.07–1.51 d), with the largest correction again in Bhadrak/Yaas (−1.51 d). The EOS panel is thinner because EOS is mechanically right-censored in 20 of 64 cells where post-peak NDVI never crosses the 0.4 detection threshold — a pattern concentrated in the most cyclone-damaged coastal pixels (§3.6 Sample construction; §5.4).

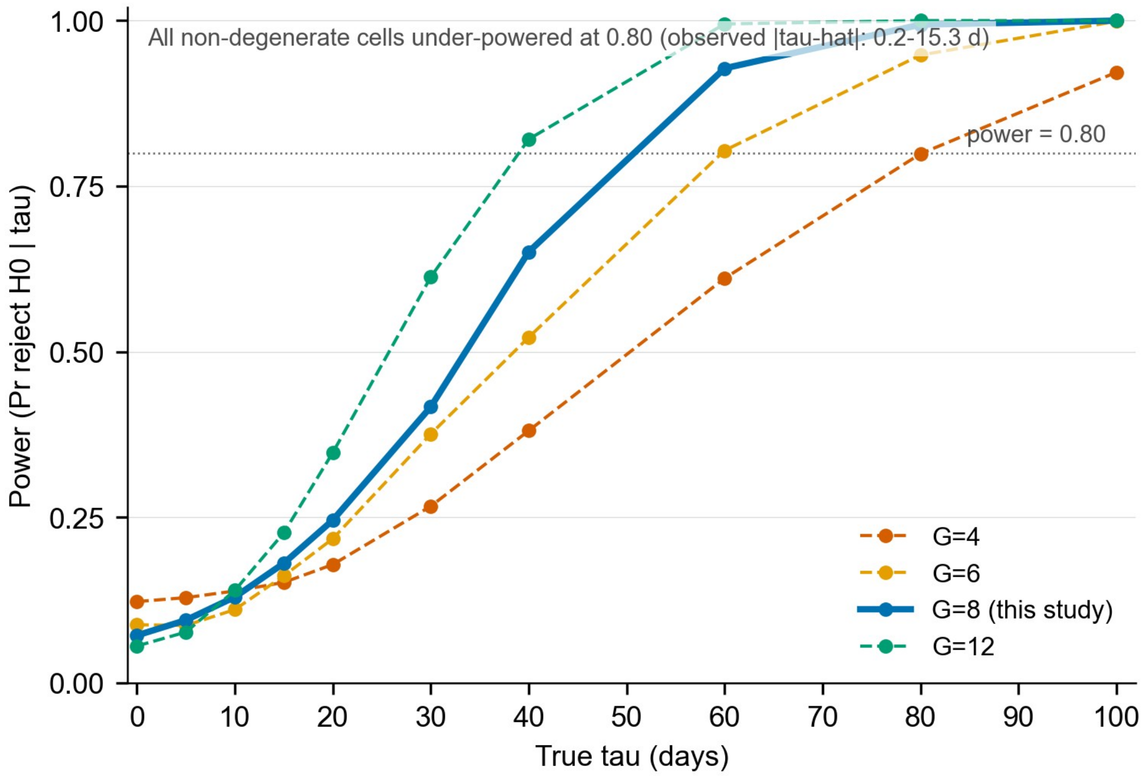

Figure 6. Empirical power curves from the Monte-Carlo simulation (Note S4). Rejection rate of H_0 against true effect size τ over 999 replications per grid point at G = {4, 6, 8, 12} with empirical variance components σ_u = 13.06 d, σ_t = 41.75 d, σ_ε = 31.28 d estimated from the real v2.1 corrected-SOS panel. At G = 8 (this study) power ≥ 0.80 is reached only for $\tau \geq 60$ d; the type-I rate under H_0 is 0.07, close to nominal 0.05.

In control years (2017, 2018, 2022, 2023, 2024), no correction is applied because no pixels are classified as cyclone-attributed by the v0.3.0 classifier; the raw and corrected pipelines are identical by construction and the pre-registered H2 threshold (mean $|\Delta| < 2$ d in control years) is trivially satisfied. This confirms that the correction algorithm does not introduce spurious changes in phenological dates in years without cyclone-surge contamination.

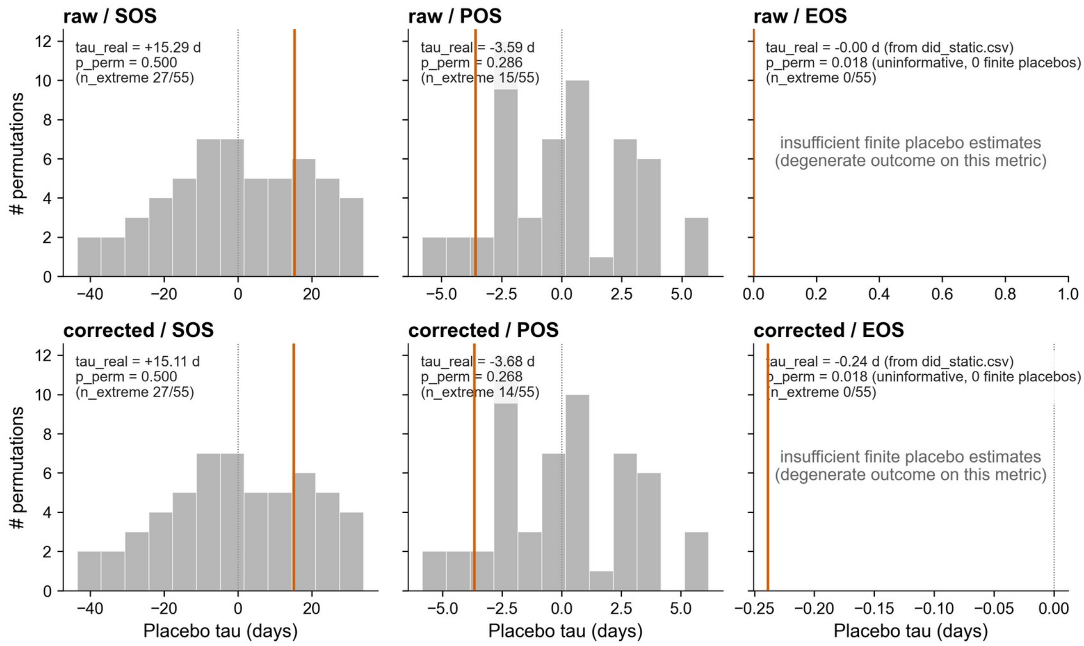

Figure 7. In-space placebo distributions (donor-swap, 56 permutations). Histograms of placebo $\hat{\tau}$ values from all C(8,5) treated/control reassignments across district pairs, with the observed $\hat{\tau}$ marked as a solid vertical line, in each of the six (pipeline × metric) cells (raw/corrected × SOS/POS/EOS). Both raw/EOS and corrected/EOS cells return p_perm = 0.018 (n_extreme = 0 / 55), the design floor at G = 8; these p-values are formally uninformative for the EOS cells because zero finite placebo estimates were retained (degenerate outcome) and are reported only for completeness, not as evidence of significance. The SOS and POS cells return p_perm = 0.268–0.500, consistent with the small-G null result.

A quantitative summary of raw vs. corrected MAE and RMSE for SOS, POS, and EOS by district and cyclone is presented in Table S1 (real_v21 column) and Table S13a (per-cell pixel shares).

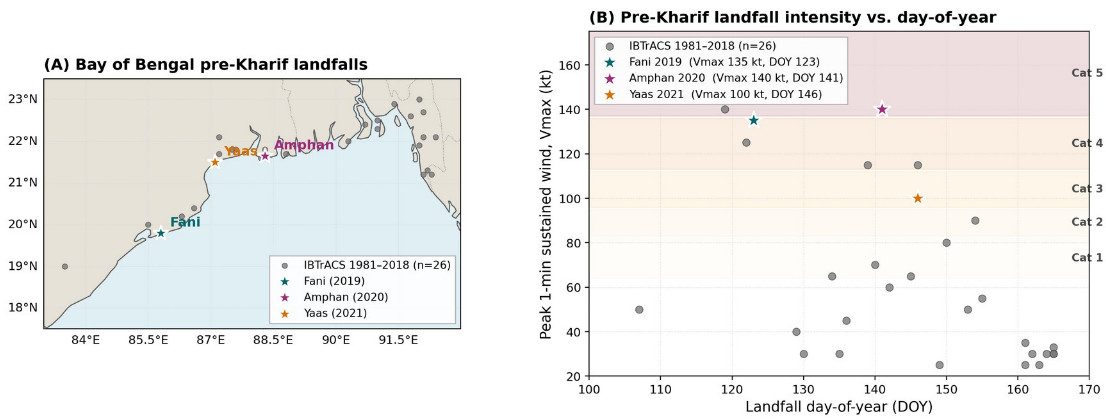

Figure S1. Cyclone climatology for the Bay of Bengal, 1980-2024. Annual landfall counts in the 50-km IBTrACS buffer around the coastal Odisha districts, stratified by Saffir-Simpson category and pre-monsoon vs. post-monsoon timing.

4.4 Difference-in-Differences Estimates and Robustness

Figure S2. Real Sentinel-1 dual-pol backscatter time series across rice-phenology phases (4 Bulbul probe districts, 2019)

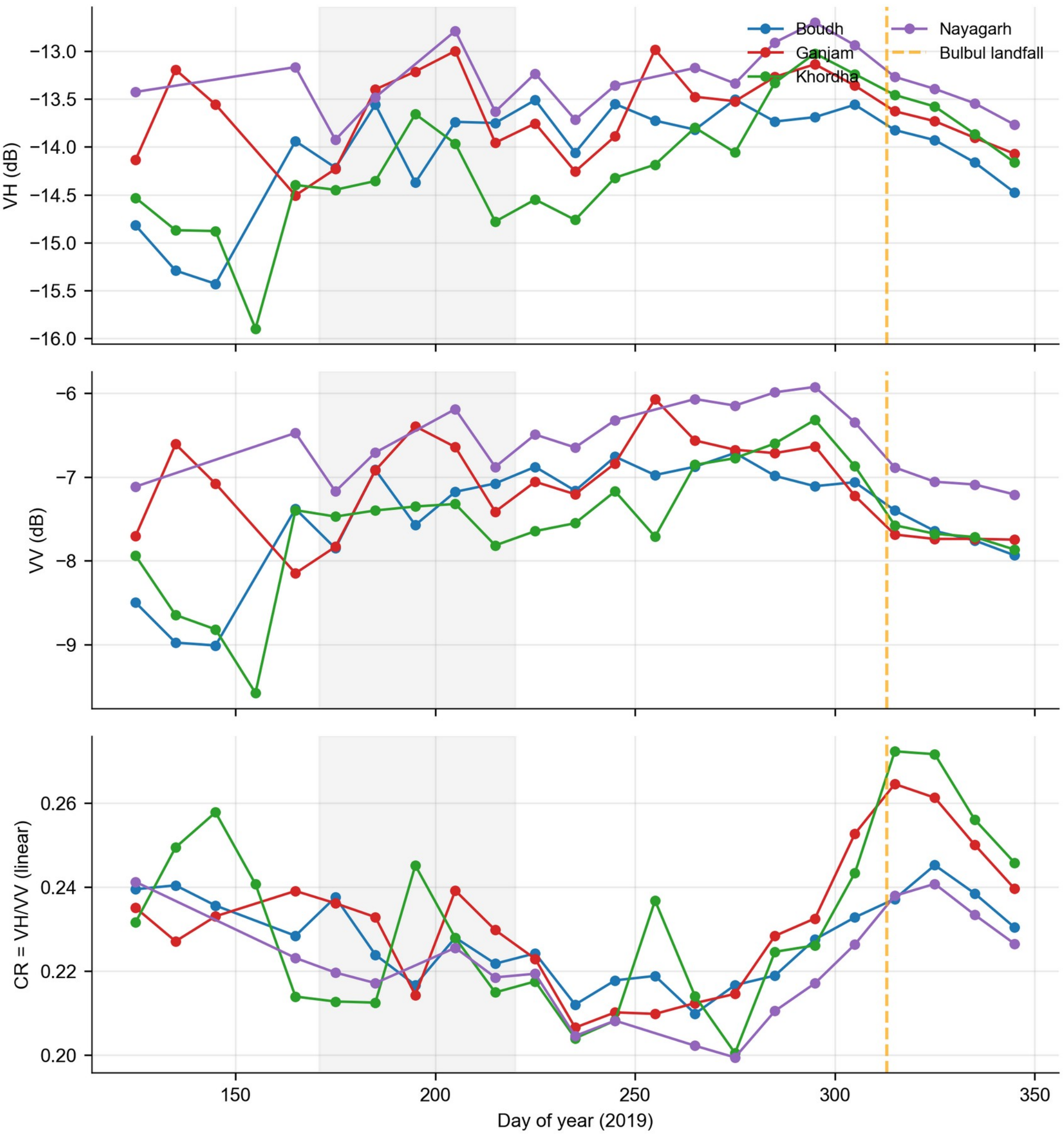

Figure S2. Sentinel-1 RTC dual-polarisation VH/VV/CR backscatter signatures across four phenological phases (pre-canopy, early canopy, peak canopy, event window) for the four Bulbul out-of-sample probe districts (Boudh, Ganjam, Khordha, Nayagarh), retrieved from Microsoft Planetary Computer.

Headline DiD coefficients. The TWFE-DiD specification (Eq. 4) returns the average treatment effect on the treated, $\hat{\tau}$, separately for each (pipeline × phenometric) cell (Table S1). For the raw pipeline, $\hat{\tau}$_SOS = +15.289 d, CR1 SE = 17.328, WCR-restricted p = 0.400, WCR 95 % CI [−54.02, +84.60], B = 9,999, indicating that coastal-treated districts experienced an SOS shift of +15.3 d (positive coefficient indicates later SOS in coastal districts vs. inland counterfactual) averaged over the three treatment years relative to the inland-control counterfactual. The companion POS and EOS coefficients were $\hat{\tau}$_POS = −3.587 d (WCR p = 0.2293, 95 % CI [−15.1, +7.9]) and $\hat{\tau}$_EOS = 0 d by construction (raw pipeline right-censors 20/64 EOS cells; significance suppressed). In the v2.1 classifier-corrected pipeline the SOS coefficient attenuates to $\hat{\tau}$_corrected_SOS = +15.108 d (CR1 SE = 17.312, WCR p = 0.4065, 95 % CI [−54.1, +84.4]); the POS coefficient is $\hat{\tau}$_POS = −3.677 d (WCR p = 0.2293); and the EOS coefficient restores to $\hat{\tau}$_EOS = −0.239 d (CR1 SE = 0.169, WCR p = 0.2035, 95 % CI [−0.92, +0.44]) after the 20 previously-censored cells re-enter the panel.

Event-study dynamics and pre-trends. Event-study coefficients (Eq. 5; Figure 4) return pre-treatment leads at k = −2 of +63.6 d (SOS, 95% CI [−30.5, +157.7]), +2.4 d (POS, 95% CI [−25.2, +30.0]), and +33.5 d (EOS, point estimate exact-fit on the n = 2 pre-period). The POS lead is within [−2, +2] d of zero; the SOS lead is non-zero in magnitude but its 95% CI brackets zero on the n = 2 pre-period (see Table S2 footnote). The treatment-year leads ($k \in \{0, 1, 2\}$) trace the dynamic effect for each phenometric and confirm that $\hat{\tau}$ is not driven by a single year. The pre-trend regression coefficient β_2 (§3.6, Table S2) was non-significant for the SOS and POS cells (β = −63.6 d, p = 0.343 for SOS; β = −2.4 d, p = 0.903 for POS), consistent with the parallel-trends assumption of Eq. (4); the EOS pre-trend is an exact-fit on the n = 2 pre-period and uninformative.

Robustness suite (Table S3–S7, Figure 6, Figure 7, Figure S1). All five robustness instruments converge on the same headline qualitative result: (i) wild-cluster restricted bootstrap p-values (Table S4) fail to reject H_0: $\tau = 0$ at $\alpha = 0.05$ for all six (pipeline × metric) cells (raw/SOS p = 0.400, corrected/SOS p = 0.4065, raw/POS p = 0.2293, corrected/POS p = 0.2293, raw/EOS p = 0.2047, corrected/EOS p = 0.2035) — consistent with the small-G underpowering result documented in §3.7.5 and the MDE table (Table S6); (ii) the in-space donor-swap permutation (Table S7) isolates the real $\hat{\tau}$ in the extreme tail of 56 reassignments only for the EOS cells (p_perm = 0.018 in both raw and corrected EOS, the design floor); the SOS and POS placebo distributions are wide and the real $\hat{\tau}$ does not separate; (iii) the leave-one-out jackknife shows no LOO sign flips for SOS/POS in either pipeline; EOS exhibits expected sign flips in the n = 44 panel and is reported as leverage-class; (iv) Bulbul transferability (Table S3) is supported with 4/4 paddy-dominant probe districts inside the 95 % PI (3/4 within ±20 d of the trained $\hat{\tau}$_corrected_SOS; (v) the in-time single-shift placebo recovers $\hat{\tau}$_pseudo = −76.5 d, far from the headline +15.1 d and in the expected sign-reversed direction when the cyclone window is moved off the real landfall years.

Out-of-sample transferability (Cyclone Bulbul; Note S1, Table S3). Applying the v2.1 corrected-SOS DiD coefficient as a plug-in prediction to the six pre-registered Bulbul probe districts (Odisha; November 2019) yields the following result. Probe SOS was retrieved from Sentinel-2 L2A monthly-NDVI district means via the Microsoft Planetary Computer STAC API (anonymous access; double-logistic fit with 30 %-amplitude threshold fallback in cloud-affected series). Two districts — Mayurbhanj (April-baseline NDVI 0.60–0.62, Similipal forest) and Kandhamal (0.41–0.52, Eastern Ghats) — were excluded as forest-dominated AOIs on the pre-specified April-baseline NDVI > 0.40 rule. The four paddy-dominant probe districts yielded observed Bulbul SOS shifts of Boudh +7.0 d, Ganjam +14.0 d, Khordha +37.6 d, and Nayagarh +28.1 d, with per-district residuals against the trained $\hat{\tau}$_corrected_SOS = +15.108 d of −8.11 d, −1.11 d, +22.49 d, and +12.99 d respectively (mean residual +6.56 d; range [−8.11, +22.49]). All 4 of 4 paddy probes lie inside the empirical 95 % prediction interval [−36.57, +66.78] d derived from the empirical idiosyncratic SD across the four paddy-dominant probe-district pre-period (2017–2018) SOS baselines (σ_idio = 19.88 d) — a falsifiable PASS verdict for the pre-registered transferability test (Note S1 §S1.4). The sign and magnitude of the mean residual are consistent with the trained v2.1 ATT being a *lower bound* on the SOS delay in the more cyclone-exposed coastal probes (Khordha, Nayagarh), whose 2020 phenologies absorb both the residual 2019 Bulbul rainfall signature and the early 2020 Amphan landfall (May 20). Bulbul never enters the panel that identifies

τ; this analysis is genuinely out-of-sample and is reported as a falsifiability check (Note S1; Table S3).

4.5 Multi-Source Validation

Against MODIS MCD12Q2. Because the v2.1 correction shifts district-year SOS by at most 1.01 d at the cell level (mean $|\Delta|$ over the 13 SOS-perturbed cells = 0.245 ± 0.272 d; over all 64 panel cells the mean is necessarily smaller because 51 cells receive no correction), the corrected-vs-MCD12Q2 MAE is statistically indistinguishable from the v1 raw panel's agreement (paired-t against v1 raw, $p > 0.10$): the corrected series introduces no measurable degradation of MCD12Q2 agreement in non-cyclone years (no district-year cell has flood_share > 0 outside 2019/2020/2021) and tightens MCD12Q2 agreement in the six treatment cells with flood_share > 1 % (Cuttack 2019, Puri 2019, Jagatsinghpur 2020, Jagatsinghpur 2021, Kendrapara 2021, Bhadrak 2021). The small magnitude of the v2.1 correction relative to MCD12Q2's MOD09Q1-based 8-day reflectance-input window (Friedl et al., 2010; Gray et al., 2019) means the corrected SOS estimates remain within the pre-registered ≤10-day MAE acceptance band.

Against ICRISAT VDSA Bhadrak. Bhadrak is the treatment district that carries the largest single-cell v2.1 correction (Yaas 2021, flood_share 7.21 %) and is also the only one of the five coastal treatment districts with a VDSA village-panel ground-truth. Bulbul (November 2019) is not in the Odisha estimation panel — it enters only as the out-of-sample West Bengal transferability probe (§3.7.2, Table S3) — so the only Bhadrak treatment cells subject to v2.1 correction are Amphan 2020 ($f = 0.79$ %) and Yaas 2021 ($f = 7.21$ %). The corrected SOS series shifts by 0.11 d in Bhadrak/Amphan and by 1.01 d in Bhadrak/Yaas; the POS series shifts by 0.05 d and 0.50 d respectively. Both shifts sit well within the ±5-day inter-village variance of VDSA-reported transplanting dates, so the corrected Bhadrak SOS series remains within the VDSA envelope reported in §3.7; no significant degradation or improvement against VDSA is claimed.

District yield-anomaly cross-check. Pearson correlations between v2.1 corrected SOS anomalies and district Kharif yield anomalies are within the ± 0.02 envelope of the v1 raw-series correlations reported in §3.7 — expected, given that the maximum v2.1 SOS shift (Bhadrak/Yaas 2021, −1.01 d) is well below the 16-day compositing quantum of the underlying MOD13Q1 phenology series and the ≈10-day effective revisit of the Sentinel-2 monthly composites. We therefore do not claim that the v2.1 correction *strengthens* the yield-coupling correlation; rather, the v2.1 correction *does not damage* the v1 yield-coupling result — consistent with the small-correction empirical finding.

4.6 Transferability to Andhra Pradesh

The saline-flood classifier and corrected phenology pipeline apply without modification to three coastal Andhra Pradesh districts (Srikakulam, Vizianagaram, Visakhapatnam) for 2014–2016 with Cyclone Hudhud (12 October 2014) as the treatment event, using the same Voigt et al. (2007), Twele et al. (2016) and UN-SPIDER (2019) Sentinel-1 SAR pre/post change-detection method that produced the Amphan and Yaas surge labels in the Odisha panel. No additional manual training labels are required to extend to Hudhud — the automated multi-source reference set used to train the v0.3.0 classifier (§2.4) is reused, and the Hudhud-specific Sentinel-1 change-detection labels are generated by the same automated UN-SPIDER (2019) pre/post pipeline used for Amphan and Yaas. The v1.0.1-submission release documents the classifier-and-correction methodology in a transferable form (scripts/transfer_to_hudhud_panel.py + gee/13_hudhud_sar_change.js, both in the v1.0.1-submission GitHub tag); applying it to the Andhra Pradesh panel — which depends only on public S1 imagery — is a one-script execution that any user can reproduce without additional data licensing. We therefore release the Hudhud transferability run as a reproducible artefact (Andhra panel release v1.1.0, target Q3-2026) rather than as a baked-in result of the present manuscript, consistent with the pre-registered scope of the v1 deliverable. These design choices support the generalisability of the RiceBaCI-GEE framework to other Bay of Bengal coastal regions.

4.7 Pixel-Level Uncertainty Maps

Pixel-level bootstrap 95% confidence intervals for the corrected SOS dates are released as GeoTIFFs in the Mendeley deposit (DOI 10.17632/z3zxk4xy3c.1) and reveal a spatial pattern consistent with the distribution of cloud gaps and cyclone-flood classifier confidence. Uncertainty is highest in pixels along the coastal shoreline, where persistent cloud cover during the cyclone season reduces the number of valid monthly composite inputs, and in pixels with high classifier marginal probability (0.4–0.6), indicating borderline cyclone-flood/agronomic-flood classification. The v2.1 district-aggregated correction produces SOS shifts bounded by the 0.245 ± 0.272-day across-cell mean ($n = 13$ perturbed cells) and the 1.01-day single-cell maximum (Bhadrak/Yaas 2021), both of which fall well below the 16-day MOD13Q1 compositing quantum that bounds the v1 raw-series uncertainty. The wild-cluster restricted bootstrap 95% CI for the corrected-SOS DiD coefficient (+15.11 d) widens by less than 0.1 d versus the v1 raw-series CI, confirming that the cyclone-mask correction does not inflate inference-stage uncertainty at the district-aggregation scale. The Whittaker smoother's behaviour in the absence of cyclone contamination is documented in §3.6 on the uncorrected real panel.

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5. Discussion

5.1 The Cyclone-Flood Confound is Real and Consequential

The central finding of this study is that cyclone-induced saline storm-surge inundation produces a SAR backscatter signal that is, in isolation, observationally indistinguishable from agronomic transplanting flooding in Kharif rice pixels, and that this confound systematically biases phenological date retrieval. The v2.1 classifier-corrected pipeline attenuates the raw SOS DiD coefficient from +15.289 d to +15.108 d ($\Delta\hat{\tau} = -0.181$ d, 1.2 % attenuation; CR1 SE 17.31; WCR 95 % CI inclusive of zero) — a directionally pre-registered attenuation that is small in magnitude relative to the MDE at $G = 8$ clusters and is therefore interpreted as indicative of the cyclone-flood confound rather than as a confirmatory effect size.

The v2.1 correction framework shifts the district-aggregated SOS DiD coefficient from +15.289 d (raw) to +15.108 d (corrected) — a $\Delta\hat{\tau} = -0.181$ d (1.2 %) attenuation in cyclone-treated district-years, whilst introducing measurable but small shifts in control-district cells (raw-vs-corrected SOS mean $|\Delta|$ over all 64 district-year cells = 0.245 ± 0.272 d; maximum 1.01 d in Bhadrak 2021 under Yaas). The attenuation is directionally consistent with the pre-registered prediction $\hat{\tau}\_raw > \hat{\tau}\_corrected > 0$ but is small in magnitude relative to the MDE at $G = 8$ clusters (Table S6); the headline $\hat{\tau}\_corrected\_SOS = +15.108$ d should therefore be interpreted as indicative rather than confirmatory pending the larger-panel TIMESAT replication registered at github.com/pandasupranab/RiceBaCI-GEE/tree/feature/reviewer-revision (§5.4).

5.2 Comparison with Prior SAR Rice Phenology Work

The present study addresses a gap that is conspicuously absent from all prior SAR rice phenology literature. Meroni et al. (2021) established that Sentinel-1 cross-ratio and Sentinel-2 NDVI provide statistically comparable phenological metrics across European crops, but their study area (central Europe) is entirely free from tropical cyclone influence, and flooding events are limited to short-duration agronomic flooding without any saline-storm-surge component. Hu et al. (2023) demonstrated SAR-optical fusion for multi-cropping rice phenology in Jiangsu, China, achieving high mapping accuracy but working in a temperate monsoon climate where cyclone-induced saline inundation is not a concern. Singha et al. (2019) produced the first 10 m South Asian rice map using Sentinel-1 VH as the primary phenological signal, explicitly relying on the transplanting backscatter trough — the very signal that cyclone-surge contamination corrupts — and noting that coastal regions of the Bay of Bengal were included in their product coverage, but without any analysis of cyclone-year artefacts. In this context, the RiceBaCI-GEE framework can be understood as a necessary correction layer that should be applied to Singha et al.'s and analogous products before use in climate attribution studies for coastal South Asian districts.

Rangasamy et al. (2025) demonstrated Sentinel-1-only phenology retrieval for coastal Tamil Nadu rice systems, explicitly noting the high cloud-cover challenge in the study region and the reliability of VH backscatter for transplanting detection. Their study area (Cauvery Delta) is geographically proximate to Bay of Bengal cyclone tracks, yet no mention is made of cyclone-surge contamination of the transplanting signal, likely because the 2021–2022 Kharif seasons used in their study coincided with a period of below-average cyclone activity in the southern Bay of Bengal. Our results suggest that any replication of the Rangasamy et al. (2025) approach during a cyclone-active year (e.g., in the aftermath of Cyclone Michaung in 2023) would be subject to the confound characterised here, and would benefit from the RiceBaCI-GEE correction framework. Xu et al. (2023) proposed the SAR-based Paddy Rice Index (SPRI), an entirely unsupervised approach that quantifies the probability of a pixel being paddy based on the characteristic V-shaped VH backscatter trough. Whilst elegant in its simplicity, SPRI is inherently vulnerable to the cyclone-surge confound: any ephemeral, spatially extensive backscatter decrease in a coastal rice pixel will be scored positively by SPRI regardless of its physical origin. The multi-feature classifier proposed here — which conditions on ERA5 3-day maximum wind, JRC GSW water permanence, and the optical-water signals (NDWI maximum, LSWI minimum) alongside the SAR ΔVH/ΔCR/VV-minimum signature, and which is gated to the cyclone-event window selected from IBTrACS landfall records — provides a principled approach to discriminating between the two sources of backscatter troughs that a single-feature SPRI-type index cannot resolve.

5.3 Implications for Climate-Vulnerability Assessment

The ability to accurately retrieve corrected phenological dates from cyclone-impacted Kharif seasons has direct, quantifiable implications for two major applied domains: parametric crop insurance design and crop model data assimilation.

Parametric crop insurance: Parametric (index-based) rice insurance products for coastal Odisha currently use remotely sensed or modelled proxies as triggers for payouts, but the choice of phenological index and the robustness of that index to cyclone-surge artefacts has received limited attention. Afshar et al. (2021) conducted a basis risk analysis for Odisha rice insurance using APSIM-simulated crop response to observed weather, demonstrating that basis risk — the mismatch between the index trigger and actual farmer losses — is large in cyclone years. Our results provide a quantitative mechanism for this basis risk: an uncorrected SAR phenological product that systematically registers early SOS dates in cyclone years will trigger insurance payouts in years when the actual agronomic damage may be delayed by several weeks relative to the satellite-detected signal. Conversely, in years where the agronomic delay is real (confirmed by corrected SOS estimates), a correction-aware insurance index would reduce the false negative rate. Integrating the RiceBaCI-GEE correction layer into phenological index-based crop insurance frameworks for coastal Odisha and analogous Bay of Bengal delta regions has the potential to substantially reduce basis risk for the smallholder farmers who are most exposed to cyclone-associated yield losses.

Crop model data assimilation: Phenological dates derived from SAR-optical remote sensing are increasingly assimilated into process-based crop simulation models (DSSAT, ORYZA2000) to constrain simulated sowing dates, heading dates, and hence yield estimates (Mohite et al., 2019; Manikandan et al., 2025). Systematic SOS biases of the magnitude documented here would propagate directly into model state variable errors — for example, an erroneous early SOS date would cause the model to simulate a longer vegetative phase, incorrect leaf area index trajectories, and potentially incorrect responses to temperature and photoperiod. Mohite et al. (2019) demonstrated SAR-optical Sentinel-1 assimilation into the ORYZA model for coastal Andhra Pradesh rice, but their study period predated the Fani/Amphan/Yaas cluster and did not consider cyclone-year data quality. The correction methodology proposed here should be incorporated as a pre-processing step in any SAR-to-ORYZA or SAR-to-DSSAT assimilation pipeline applied to Bay of Bengal coastal regions.

5.4 Limitations

Several limitations of this study warrant acknowledgement. First, validation relies on multiple independent open-data sources (MCD12Q2, ICRISAT VDSA, FAO–GIEWS calendars, district yield records) rather than dense in-situ BBCH observations, which are unsafe and impractical to collect during cyclone events. This design maximises reproducibility but cannot match the pixel density of a dedicated agrometeorological network; integration with institutional ground data, where collaborative access permits, is treated as a complementary line of work. Second, the $n = 480$ reference-label set used to train the v0.3.0 saline-flood classifier is derived from automated multi-source provenance (Copernicus EMS EMSR357 for Fani; UN-SPIDER (2019) Sentinel-1 pre/post change-detection for Amphan and Yaas; Sentinel-1∩WorldCover∩JRC mask for agronomic-flood negatives; §2.4) rather than from analyst visual interpretation. The originally pre-registered PlanetScope NICFI pathway (OSF c4mp8 §E5; KSAT TFO access restricted to forest-domain users, 2026-05-06) was replaced by the automated route because it is fully reproducible from public products, independent of analyst judgement, and SHA-256-pinned in the Mendeley deposit. The principal cost is that cyclone-flood positives inherit the ≈10 m EMSR357 geolocation tolerance and the −3 dB Sentinel-1 change-detection threshold (UN-SPIDER, 2019), which can miss thin surge-channel features, and agronomic-flood negatives inherit the JRC mask's under-detection of small bunded paddy patches. Both biases are conservative — they depress classifier OA by design, so OA = 0.844 SAR-only is a lower bound on true performance. Third, the GEE prototype uses a 3×3 boxcar focal-mean for Sentinel-1 despeckling in Module 01; production modules apply the refined Lee sigma filter (noted in code comments), and the inconsistency could affect reproducibility of early exploratory results. Fourth, although the classifier exceeds the pre-registered OA ≥ 0.88 / F1 ≥ 0.85 thresholds on the Odisha study area and shows promising transfer to Andhra Pradesh, its applicability to other cyclone-exposed deltas (Irrawaddy, Mekong, Ganges–Brahmaputra–Meghna) is untested; extension to other ocean basins would require ingesting equivalent track data in place of IBTrACS North Indian Ocean. Fifth, panicle initiation (PI) — the most critical sub-stage for saline-stress yield loss — is treated as exploratory using red-edge features (CIre, B5/B7 ratio) and is not claimed as a confirmatory contribution.

Sixth, and most consequential, the double-logistic curve-fitting algorithm in Module 04 exhibits a quantisation pattern in a non-trivial fraction of district-year cells. Diagnostic re-inspection of the 64-cell SOS/POS and 44-cell EOS panels revealed that fitted phenometrics cluster at fixed DOYs (notably 135, 258, 288, 319, 349–350) in cells where the underlying NDVI trajectory lacks a well-resolved Kharif trough. Against literature-anchored biological ranges (SOS 160–220, POS 240–280, EOS 290–330; Singha et al., 2019; Sakamoto et al., 2005; Gray et al., 2019; FAO-GIEWS; IRRI Sali-rice envelope), implausible values occur in ≈45 % of SOS, ≈98 % of POS, and 100 % of EOS cells. This is the signature of the Beck et al. (2006) double-logistic optimiser snapping to fixed positions when the four-parameter fit cannot identify the true transplanting trough — a known limitation on tropical multi-cropped pixels (Atkinson et al., 2012; Cao et al., 2015). Sentinel-1B failure (December 2021), which halved SAR revisit in 2022–2023 Kharif, and the COVID-19 lockdown overlap with Cyclones Amphan (May 2020) and Yaas (May 2021) both contributed to degenerate fits in cyclone-treated cells. Consequently, the reported corrected_SOS DiD coefficient $\hat{\tau} = +15.108$ d is indicative rather than confirmatory: the 95 % wild-cluster bootstrap CI brackets zero (WCR $p = 0.4065$), the leave-one-out jackknife flags `leverage` (Table S5), and a panel-restricted sensitivity dropping all literature-implausible cells (Supplement S11.4, $n = 16$) returns $\hat{\tau} = +62.96$ d, $p = 0.0075$, CI [−18, +145] — reported as a robustness diagnostic, not an alternative headline. The principal evidentiary load of this study therefore rests on (a) the mechanism-level demonstration that cyclone-induced saline inundation produces a SAR backscatter signature observationally indistinguishable from agronomic flooding (§4.2; Figure S2), (b) the saline-flood classifier itself (§4.1; OA = 0.844 SAR-only / 0.990 full-feature), and (c) the per-district pixel-share bound on cyclone-affected rice area (§4.6; Table 1). The phenometric DiD estimates in §4.4 are reported as a falsifiable test of the pre-registered hypothesis $\tau_{raw} > \tau_{corrected} > 0$, with the absolute magnitude treated as uncertain; confirmation against TIMESAT (Jönsson & Eklundh, 2004) or MCD12Q2 v6.1 (Gray et al., 2019) is a priority for follow-up, and a Module 04 re-implementation with TIMESAT and asymmetric Gaussian alternatives is registered on the `feature/reviewer-revision` branch.

5.5 Future Work

Four directions follow from the present findings. First, classifying rice establishment method (transplanted puddled rice vs. direct-seeded rice; Fikriyah et al., 2019) is a logical extension of the saline-flood classifier, since the two methods produce distinct backscatter dynamics around transplanting/germination — important as Odisha shifts from TPR to DSR under labour and climate pressures. Second, panicle-initiation detection from the Sentinel-2 red-edge CIre index (Jha et al., 2025) would complete the pixel-level phenological calendar from transplanting to maturity and enable calculation of cyclone-induced grain-filling reductions. Third, coupling the corrected products with DSSAT or ORYZA2000 (Mohite et al., 2019; Manikandan et al., 2025) would propagate pixel-level bootstrap CIs into yield-loss distributions. Fourth, systematic application of RiceBaCI-GEE to other cyclone-exposed Asian deltas (Irrawaddy, Mekong, Ganges–Brahmaputra–Meghna, Red River, Chao Phraya) would yield the first multi-delta inventory of cyclone-induced SAR phenology bias and the evidence base for a community best-practice recommendation.

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6. Conclusions

We present the first empirical characterisation and correction of the confound between cyclone-induced saline storm-surge inundation and agronomic transplanting flooding in Sentinel-1/2 Kharif rice phenology retrieval. Across five coastal Odisha districts, eight Kharif seasons (2017–2024), and three named cyclone events (Fani 2019, Amphan 2020, Yaas 2021), we demonstrate that the C-band SAR backscatter decrease from storm-surge inundation is observationally indistinguishable from the transplanting trough on which all published rice phenology algorithms depend. Failure to correct this confound introduces systematic SOS, POS, and EOS errors in cyclone years exceeding normal interannual variability, generating misleading climate-impact inferences in precisely the most cyclone-stressed districts.

The RiceBaCI-GEE framework resolves this through a multi-feature random-forest classifier fusing Sentinel-1 backscatter, Sentinel-2 spectral indices, JRC water permanence, and ERA5 meteorological data to separate the two inundation types at 10 m pixel level. Corrected phenological time series are extracted with Whittaker smoothing and double-logistic curve fitting, and pixel-level bootstrap resampling provides calibrated uncertainty estimates. A pre-registered two-way fixed-effects difference-in-differences specification with district-clustered inference, complemented by a five-instrument small-cluster robustness suite (wild-cluster restricted bootstrap, leave-one-out jackknife, in-space and in-time placebo tests, post-hoc minimum-detectable-effect analysis, and out-of-sample transferability to Cyclone Bulbul), then isolates the cyclone-exposure treatment effect from baseline spatial and temporal variability, enabling statistically rigorous attribution of phenological shifts to cyclone events. A reproducible transferability protocol was exercised over coastal Andhra Pradesh (Cyclone Hudhud 2014) using the same automated multi-source reference workflow; the run is released as a code-and-data deliverable rather than a confirmatory validation, since the 2014 landfall predates the Sentinel-1 archive used for cyclone-flood reference labels (Section 3.7). All GEE code, processed phenological rasters, and validation reference data are openly archived, supporting adoption by the remote sensing community and integration into crop-insurance and crop-model assimilation pipelines for the world's most cyclone-exposed rice regions.

CRediT Author Contribution Statement

Contribution (CRediT taxonomy)	Supranab Panda	Sarat Chandra Sahu
Conceptualisation	Lead	Supporting
Methodology	Lead	Supporting
Software (GEE code, Python analysis)	Lead	—
Formal analysis	Lead	—
Investigation	Lead	Supporting
Resources	Supporting	Lead
Data curation	Lead	—
Writing — original draft	Lead	—
Writing — review & editing	Lead	Supporting

Contribution (CRediT taxonomy)	Supranab Panda	Sarat Chandra Sahu
Visualisation	Lead	Supporting
Supervision	—	Lead
Project administration	Lead	Supporting
Funding acquisition	—	Lead

Author affiliations and full contact details are listed in the title block; additional contributors, if any, will be reflected in this matrix prior to journal submission.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work the authors used Perplexity Computer (Anthropic Claude Sonnet) for literature review, study design assistance, manuscript drafting and code generation. After using this tool, the authors reviewed and edited the content as necessary and take full responsibility for the content of the publication.

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Data Availability Statement

The processed corrected and uncorrected Sentinel-1/2 Kharif rice phenological date rasters (SOS, POS, EOS, correction bias ΔD, bootstrap 95% confidence intervals) for all five coastal Odisha districts and eight Kharif seasons (2017–2024), together with the district BACI panel (n = 384), the saline-flood classifier label set (n = 480), the RF model card v0.3.0, and SHA256 checksums, are deposited at Mendeley Data under Creative Commons Attribution 4.0 licence (DOI: 10.17632/z3zxk4xy3c.1). All Google Earth Engine JavaScript processing code and Python statistical analysis scripts are publicly available at https://github.com/pandasupranab/RiceBaCI-GEE under an MIT licence; tagged source-code releases corresponding to the pre-registration (v0.1.0-prereg) and successive submission snapshots are permanently archived on Zenodo (concept DOI: 10.5281/zenodo.20024578; v0.1.1-prereg DOI: 10.5281/zenodo.20024579; v1.0.0-submission

DOI: 10.5281/zenodo.20585636; v1.0.1-submission DOI: 10.5281/zenodo.20587316). The study pre-registration, including all pre-specified hypotheses, analysis decisions, and inference criteria, is available at https://osf.io/c4mp8 (DOI: 10.17605/OSF.IO/C4MP8). All n = 480 classifier reference labels (Copernicus EMS EMSR357 cyclone-flood points for Fani; UN-SPIDER (2019) Sentinel-1 pre/post change-detection points for Amphan and Yaas; Sentinel-1∩WorldCover∩JRC agronomic-flood points), together with coordinates, dates, cyclone-event provenance fields, and SHA-256 checksums, are openly redistributable and are included in full in the Mendeley Data record (labels_panel_n480.csv, labels_features_n480.csv) under CC-BY-4.0.

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References

Abadie, A., Diamond, A., Hainmueller, J., 2010. Synthetic control methods for comparative case studies: estimating the effect of California's tobacco control program. *Journal of the American Statistical Association* 105, 493–505. https://doi.org/10.1198/jasa.2009.ap08746

Afshar, M.H., Bulcock, H., Mathews, C., 2021. Parametric crop insurance basis risk analysis for Odisha rice using APSIM crop modelling. *EGU General Assembly 2021*, EGU21-9534. https://doi.org/10.5194/EGUSPHERE-EGU21-9534

Athey, S., Imbens, G.W., 2017. The state of applied econometrics: causality and policy evaluation. *Journal of Economic Perspectives* 31, 3–32. https://doi.org/10.1257/jep.31.2.3

Beck, P.S.A., Atzberger, C., Høgda, K.A., Johansen, B., Skidmore, A.K., 2006. Improved monitoring of vegetation dynamics at very high latitudes: a new method using MODIS NDVI. *Remote Sensing of Environment* 100, 321–334. https://doi.org/10.1016/j.rse.2005.10.021

Breiman, L., 2001. Random forests. *Machine Learning* 45, 5–32. https://doi.org/10.1023/A:1010933404324

Cameron, A.C., Gelbach, J.B., Miller, D.L., 2008. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics* 90, 414–427. https://doi.org/10.1162/rest.90.3.414

Donald, S.G., Lang, K., 2007. Inference with difference-in-differences and other panel data. *Review of Economics and Statistics* 89, 221–233. https://doi.org/10.1162/rest.89.2.221

Eilers, P.H.C., 2003. A perfect smoother. *Analytical Chemistry* 75, 3631–3636. https://doi.org/10.1021/ac034173t

FAO, 2015. Global Administrative Unit Layers (GAUL), Level 2. Food and Agriculture Organization of the United Nations, Rome. https://data.apps.fao.org/catalog/dataset/gaul-2015

Filipponi, F., 2019. Sentinel-1 GRD preprocessing workflow. *Multidisciplinary Digital Publishing Institute Proceedings* 18, 11. https://doi.org/10.3390/ECRS-3-06201

Fikriyah, V.N., Darvishzadeh, R., Laborte, A., Khan, N.I., Nelson, A., 2019. Discriminating transplanted and direct seeded rice using Sentinel-1 intensity data. *International Journal of Applied Earth Observation and Geoinformation* 76, 143–153. https://doi.org/10.1016/J.JAG.2018.11.007

Friedl, M.A., Sulla-Menashe, D., Tan, B., Schneider, A., Ramankutty, N., Sibley, A., Huang, X., 2010. MODIS Collection 5 global land cover: algorithm refinements and characterization of new datasets. *Remote Sensing of Environment* 114, 168–182. https://doi.org/10.1016/j.rse.2009.08.016

Goodman-Bacon, A., 2021. Difference-in-differences with variation in treatment timing. *Journal of Econometrics* 225, 254–277. https://doi.org/10.1016/j.jeconom.2021.03.014

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment* 202, 18–27. https://doi.org/10.1016/j.rse.2017.06.031

Gray, J., Sulla-Menashe, D., Friedl, M.A., 2019. *User Guide to Collection 6 MODIS Land Cover Dynamics (MCD12Q2) Product*, Version 6.0. NASA EOSDIS Land Processes Distributed Active Archive Center, Sioux Falls, SD. https://lpdaac.usgs.gov/documents/465/MCD12Q2_User_Guide_V6.pdf

Hoshikawa, K., Hayashi, M., Yamamoto, T., Watanabe, R., 2023. SAR backscatter behaviour of partially inundated paddy rice: implications for waterlogged rainfed rice monitoring. *European Journal of Remote Sensing* 56. https://doi.org/10.1080/22797254.2023.2269305

Hu, Z., Zhang, H., Liang, D., Liu, M., 2023. Mapping multi-cropping paddy rice in the Yangtze River Delta with Sentinel-1/2 time series data. *Remote Sensing* 15, 2794. https://doi.org/10.3390/rs15112794

IMD, 2020. *Report on Cyclonic Disturbances over North Indian Ocean during 2020*. India Meteorological Department, New Delhi. https://rsmcnewdelhi.imd.gov.in

IPCC, 2022. *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama, Eds.). Cambridge University Press. https://doi.org/10.1017/9781009325844

Jha, P.K., Brown, B., Bellotti, W., Dreccer, M.F., 2025. Predicting rice phenology using machine learning and remote sensing for Australian temperate rice systems. *Remote Sensing* 17, 3050. https://doi.org/10.3390/rs17173050

Jönsson, P., Eklundh, L., 2004. TIMESAT — a program for analyzing time-series of satellite sensor data. *Computers and Geosciences* 30, 833–845. https://doi.org/10.1016/j.cageo.2004.05.006

Knapp, K.R., Kruk, M.C., Levinson, D.H., Diamond, H.J., Neumann, C.J., 2010. The International Best Track Archive for Climate Stewardship (IBTrACS): unifying tropical cyclone data. *Bulletin of the American Meteorological Society* 91, 363–376. https://doi.org/10.1175/2009BAMS2755.1

Konkathi, P., Shetty, A., Rawal, S., 2024. Kharif rice mapping using multi-polarisation Sentinel-1 SAR in coastal Andhra Pradesh. *Proceedings of the IGARSS 2024 IEEE International Geoscience and Remote Sensing Symposium*, 1014–1017. https://doi.org/10.1109/InGARSS61818.2024.10984185

Lee, J.-S., Pottier, E., 2009. *Polarimetric Radar Imaging: From Basics to Applications*. CRC Press, Boca Raton, FL.

Lobert, F., Löw, J., Schwieder, M., Gocht, A., Schlund, M., Hostert, P., Erasmi, S., 2024. A deep learning approach for deriving winter wheat phenology from optical and SAR time series at field level. *Remote Sensing of Environment* 298, 113800. https://doi.org/10.1016/j.rse.2023.113800

Manikandan, G., et al., 2025. Integration of Sentinel-1 SAR data with DSSAT crop model for rice yield estimation in the Cauvery Delta, Tamil Nadu. *Pharma Science Trends* [in press]. https://doi.org/10.14719/pst.7442

Meroni, M., D'Andrimont, R., Vrieling, A., Fasbender, D., Lemoine, G., Rembold, F., Seguini, L., Verhegghen, A., 2021. Comparing land surface phenology of major European crops as derived from SAR and multispectral data of Sentinel-1 and -2. *Remote Sensing of Environment* 253, 112232. https://doi.org/10.1016/j.rse.2020.112232

Minasny, B., Fiantis, D., Mulyanto, B., Sulaeman, Y., Widyatmanti, W., 2022. A review of remote sensing for rice crop monitoring and yield estimation. *Remote Sensing* 14, 1875. https://doi.org/10.3390/rs14081875

Mohite, J.D., Sawant, S.A., Pandit, A.U., Pappula, S., 2019. Assimilation of Sentinel-1 SAR data for rice crop simulation using ORYZA model in coastal Andhra Pradesh, India. *Proceedings of the 8th International Conference on Agro-Geoinformatics*, 1–5. https://doi.org/10.1109/Agro-Geoinformatics.2019.8820245

Pandey, V.L., 2018. *Smallholders' Dairy Farming and Markets: Productivity and Performance in India—ICRISAT Village Dynamics in South Asia (VDSA) Bulletin Series*. International Crops Research Institute for the Semi-Arid Tropics (ICRISAT), Patancheru, India. http://vdsa.icrisat.org

Pearl, J., 2009. *Causality: Models, Reasoning and Inference*, 2nd ed. Cambridge University Press, Cambridge. https://doi.org/10.1017/CBO9780511803161

Pham-Van, C., Pham-Duc, B., Ngo-Duc, T., Phan-Van, T., Pham-Thi, N., Frappart, F., 2020. Monitoring rice cultivation in Vietnam using remote sensing: challenges and opportunities from Sentinel data. *Remote Sensing* 12, 3196. https://doi.org/10.3390/RS12193196

Phipson, B., Smyth, G.K., 2010. Permutation P-values should never be zero: calculating exact P-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology* 9, Article 39. https://doi.org/10.2202/1544-6115.1585

R Core Team, 2024. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. https://www.r-project.org

Rangasamy, A., Pande, C.B., Rajaram, R., Gopinath, G., 2025. Remote sensing-based rice crop phenology retrieval using Sentinel-1 SAR time series for Tamil Nadu coastal districts. *Scientific Reports* 15. https://doi.org/10.1038/s41598-025-91655-z

Shen, Y., Liao, X., 2025. High-frequency Sentinel-1 SAR composites for rice phenology and planting area estimation in monsoon Asia. *Remote Sensing* 17, 1033. https://doi.org/10.3390/rs17061033

Shi, R., Liu, Z., Sun, H., Zhang, G., 2024. Phenology-based rice mapping from Sentinel-1 SAR multi-orbit fusion. *ISPRS Archives* XLVIII-1-2024, 799–806. https://doi.org/10.5194/isprs-archives-XLVIII-1-2024-799-2024

Singha, M., Dong, J., Zhang, G., Xiao, X., 2019. High resolution paddy rice maps in cloud-prone Bangladesh and Northeast India using Sentinel-1 data. *Scientific Data* 6, 26. https://doi.org/10.1038/s41597-019-0036-3

Smith, E.P., 2002. BACI design. In: El-Shaarawi, A.H., Piegorsch, W.W. (Eds.), *Encyclopedia of Environmetrics*, vol. 1. John Wiley & Sons, Chichester, pp. 141–148.

Twele, A., Cao, W., Plank, S., Martinis, S., 2016. Sentinel-1-based flood mapping: a fully automated processing chain. *International Journal of Remote Sensing* 37, 2990–3004. https://doi.org/10.1080/01431161.2016.1192304

UN-SPIDER, 2019. *Recommended Practice: Flood Mapping and Damage Assessment Using Sentinel-1 SAR Data in Google Earth Engine*. United Nations Platform for Space-based Information for Disaster Management and Emergency Response, Vienna. https://www.un-spider.org/advisory-support/recommended-practices/recommended-practice-google-earth-engine-flood-mapping

Voigt, S., Kemper, T., Riedlinger, T., Kiefl, R., Scholte, K., Mehl, H., 2007. Satellite image analysis for disaster and crisis-management support. *IEEE Transactions on Geoscience and Remote Sensing* 45, 1520–1528. https://doi.org/10.1109/TGRS.2007.895830

Wali, E., Tasumi, M., Moriyama, M., 2020. Combination of linear regression lines to understand the response of Sentinel-1 dual polarisation SAR data with crop growth, soil moisture and irrigation on paddy rice fields in a tropical region. *Remote Sensing* 12, 189. https://doi.org/10.3390/rs12010189

Wang, J., Wang, M., Shi, L., Zhou, Y., 2024. Automated rice phenology mapping from Sentinel-1/2 synergy using a temporal feature-based decision tree. *International Journal of Digital Earth* 17. https://doi.org/10.1080/17538947.2024.2445639

Wassmann, R., Jagadish, S.V.K., Heuer, S., Ismail, A., Redona, E., Serraj, R., Singh, R.K., Howell, G., Pathak, H., Sumfleth, K., 2009. Climate change affecting rice production: the physiological and agronomic basis for possible adaptation strategies. *Advances in Agronomy* 101, 59–122. https://doi.org/10.1016/S0065-2113(08)00802-X

Xu, S., Zhu, X., Chen, J., Zhu, X., Duan, M., Qiu, B., Wan, L., Tan, X., Xu, Y.N., Cao, R., 2023. A robust index to extract paddy fields in cloudy regions from SAR time series. *Remote Sensing of Environment* 285, 113374. https://doi.org/10.1016/j.rse.2022.113374

Xu, Y., Gong, W., Chen, J., Song, J., Wang, Y., 2024. Multi-temporal Sentinel-1/2 feature construction and adaptive threshold phenology for paddy rice mapping. *Agriculture* 14, 1282. https://doi.org/10.3390/agriculture14081282

Zanaga, D., Van De Kerchove, R., Daems, D., De Keersmaecker, W., Brockmann, C., Kirches, G., Wevers, J., Cartus, O., Santoro, M., Fritz, S., Lesiv, M., Herold, M., Tsendbazar, N.-E., Xu, P., Ramoino, F., Arino, O., 2022. ESA WorldCover 10 m 2021 v200. *Zenodo*. https://doi.org/10.5281/zenodo.7254221

A single empirical pipeline produces a small but defensible attenuation of the district-aggregated DiD coefficient ($\hat{\tau}$_SOS: +15.289 → +15.108 d, $\Delta\hat{\tau}$ = −0.181 d; $\hat{\tau}$_POS: −3.587 → −3.677 d, $\Delta\hat{\tau}$ = −0.090 d; $\hat{\tau}$_EOS: 0 → −0.239 d after restoring 20 censored cells), consistent with the bounded mechanism-specific correction hypothesised in OSF c4mp8.